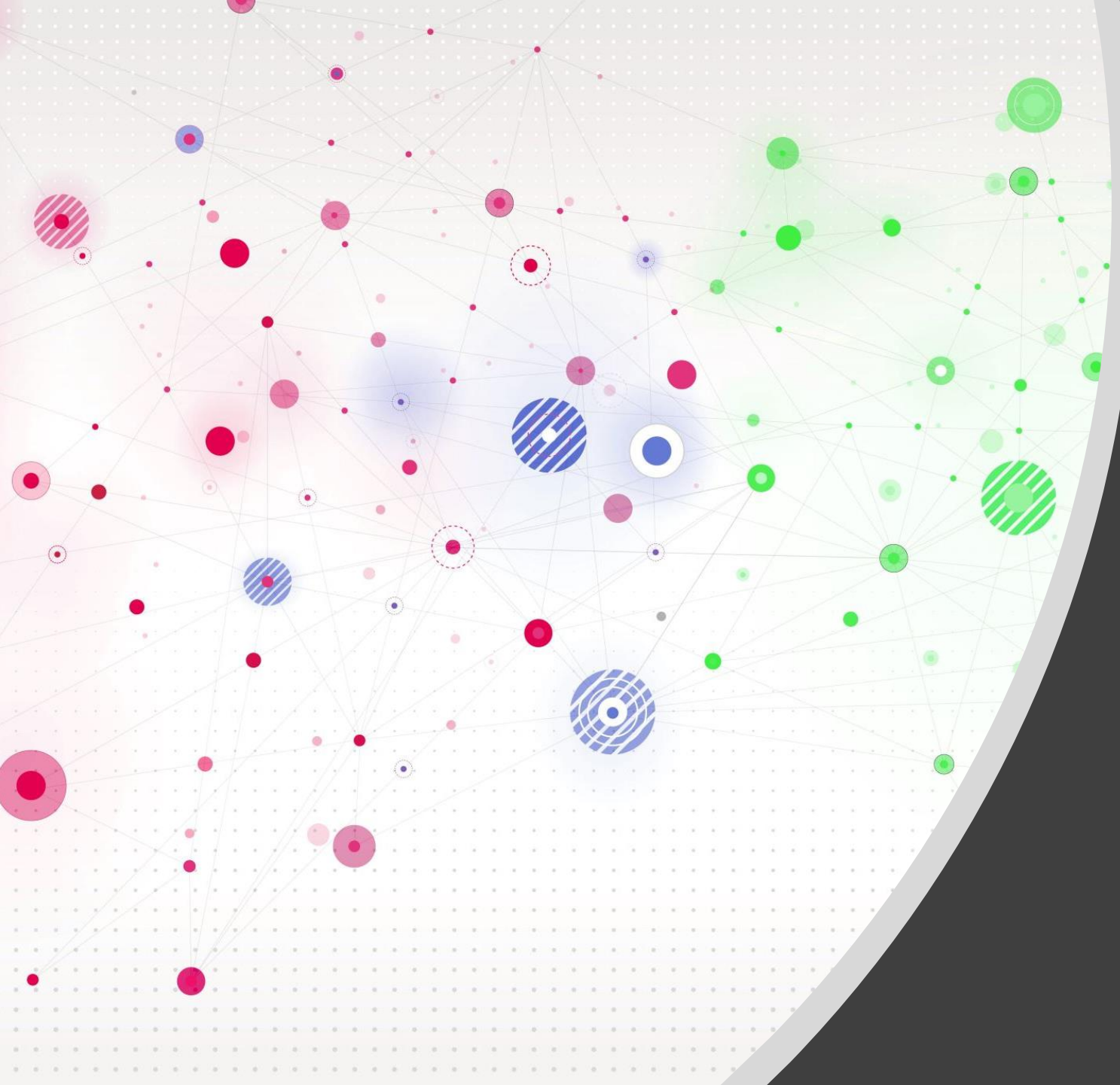


Clustering in Machine Learning

Arockiaraj S



Day-1 Clustering

Clustering

- When a Chief Marketing Officer of a company tells you, “Help me understand our customers better so that we can market our products to them in a better manner!”

Addressing individual customer – shopping history ???



Group of customers – shopping history ???



Outline

- Overview
- Learning from unclassified data
- Non-Hierarchical Clustering – k-means partitional clustering
- Hierarchical - Agglomerative Clustering.
- Expectation maximization (EM) for soft clustering.
- Semi-supervised learning with EM using labeled and unlabled data.

Overview

- Clustering is the task of dividing the data points into a number of groups such that data points in a group are more similar than those in other groups.
- The method of identifying similar groups of data in a dataset is called clustering
- It is one of the most popular techniques in data science

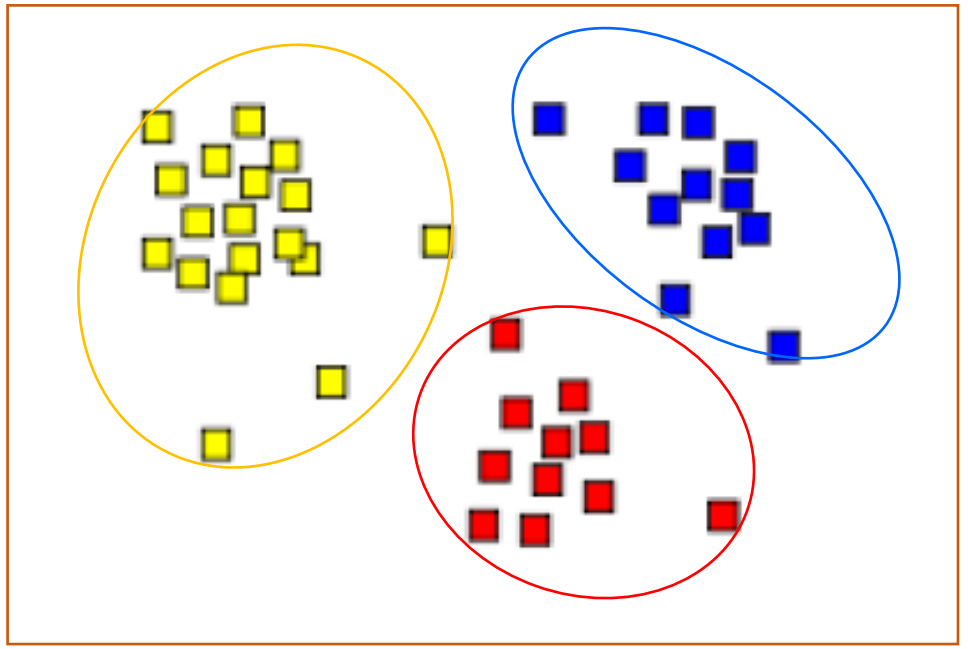
What is clustering?

The organization of unlabeled data into similarity groups called clusters.

Data items within a cluster are "*similar*"

Data items between clusters are "*dissimilar*"

- Clustering is a unsupervised learning method
- Used for statistical data analysis in many fields.



Types of Clustering

Clustering can be divided into two subgroups

- **Hard Clustering:**
  - In hard clustering, each data point may belongs to a cluster completely or not.
  - For example, each customer is put into one group out of the 10 groups.

- **Soft Clustering**:
  - In soft clustering, instead of putting each data point into a separate cluster, a probability or likelihood of that data point to be in those clusters is assigned.
  - For example, each costumer is assigned a probability to be in either of 10 clusters of the retail store.

Types of clustering algorithms

- **Connectivity models:**
  - These models are based on the notion that the data points closer in data space exhibit more similarity to each other than the data points lying farther away.
  - Two approached in these models
    - First approach: start with classifying all data points into separate clusters & then aggregating them as the distance decreases.
    - Second approach: all data points are classified as a single cluster and then partitioned as the distance increases.
    - Lacks scalability for handling big datasets.
    - Example: Hierarchical clustering algorithm and its variants.

In fact, there are more than 100 clustering algorithms available. Here we discuss only few of the algorithms are used popularly used:

Types of clustering algorithms

- **Centroid models:**
  - These are iterative clustering algorithms in which the notion of similarity is derived by the closeness of a data point to the centroid of the clusters.
  - K-Means clustering algorithm is a popular algorithm that falls into this category.
  - In these models, the no. of clusters required need to be mentioned beforehand, which makes it important to have prior knowledge of the dataset.
  - These models run iteratively to find the local optima.

Distribution models

- These clustering models are based on the notion of how probable is it that all data points in the cluster belong to the same distribution (For example: Normal, Gaussian).
- These models often suffer from overfitting.
- A popular example of these models is Expectation-maximization algorithm which uses multivariate normal distributions.

Density Models

- These methods consider the clusters as the dense region having some similarity and different from the lower dense region of the space.

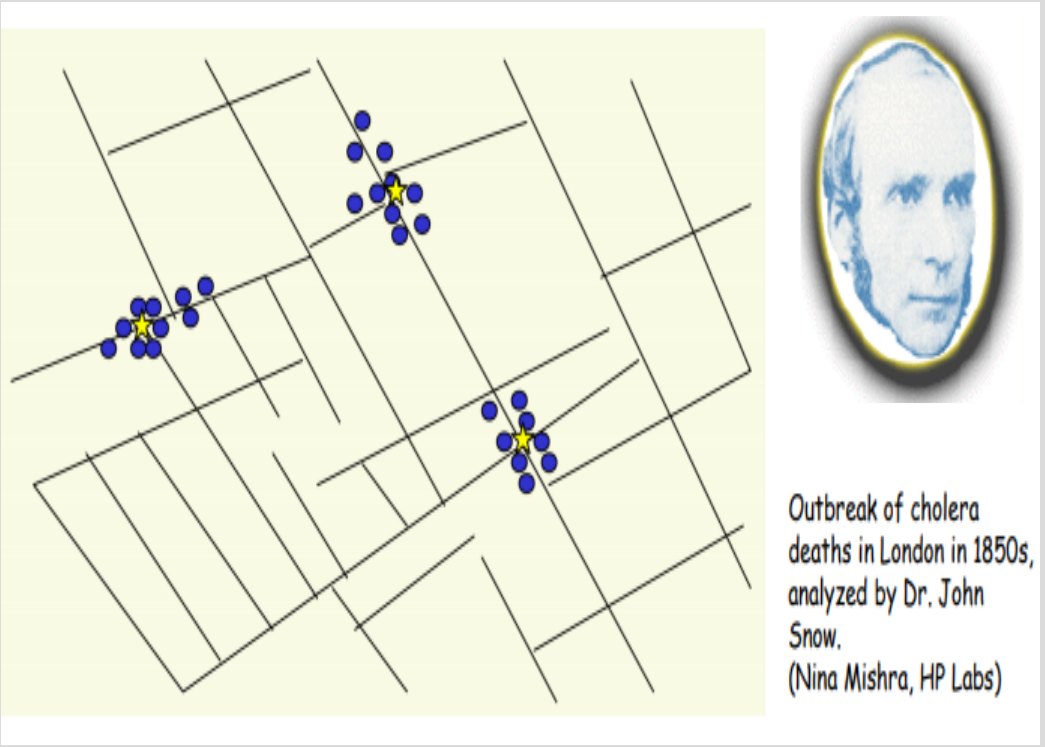
- These methods have good accuracy and ability to merge two clusters.

- Example
  - *DBSCAN (Density-Based Spatial Clustering of Applications with Noise)*
  - *OPTICS (Ordering Points to Identify Clustering Structure)*

Applications of Clustering

Historic application of clustering (First?)

- John Snow, a London physician plotted the location of cholera deaths on a map during an outbreak in the 1850s.
- The locations indicated that cases were clustered around certain intersections where there were polluted wells, thus showing both the problem and solution.



Outbreak of cholera deaths in London in 1850s, analyzed by Dr. John Snow.
(Nina Mishra, HP Labs)

Applications of Clustering in Different Fields

Marketing: It can be used to characterize & discover customer segments for marketing purposes.

Biology: It can be used for classification among different species of plants and animals.

Libraries: It is used in clustering different books on the basis of topics and information.

Insurance: It is used to acknowledge the customers, their policies and identifying the frauds.

City Planning: It is used to make groups of houses and to study their values based on their geographical locations and other factors present.

Earthquake studies: By learning the earthquake affected areas we can determine the dangerous zones.

Applications of Clustering

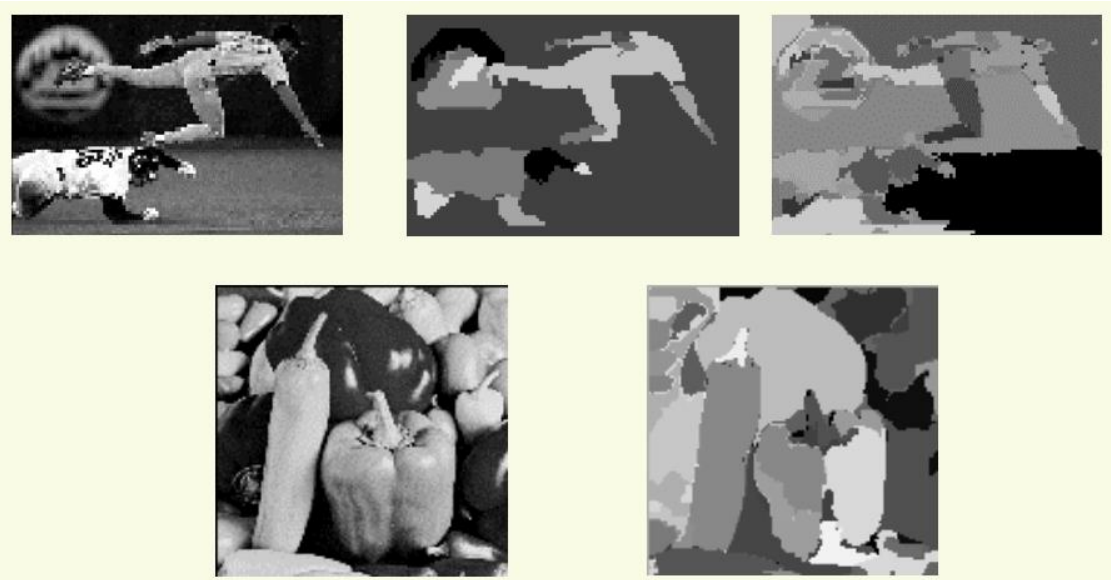
Astronomy:

- SkyCat: Clustered $2x10^9$ sky objects into stars, galaxies, quasars, etc based on radiation emitted in different spectrum bands.



Image segmentation:

- Find interesting "objects" in images to focus attention



From: Image Segmentation by Nested Cuts, O. Veksler, CVPR2000

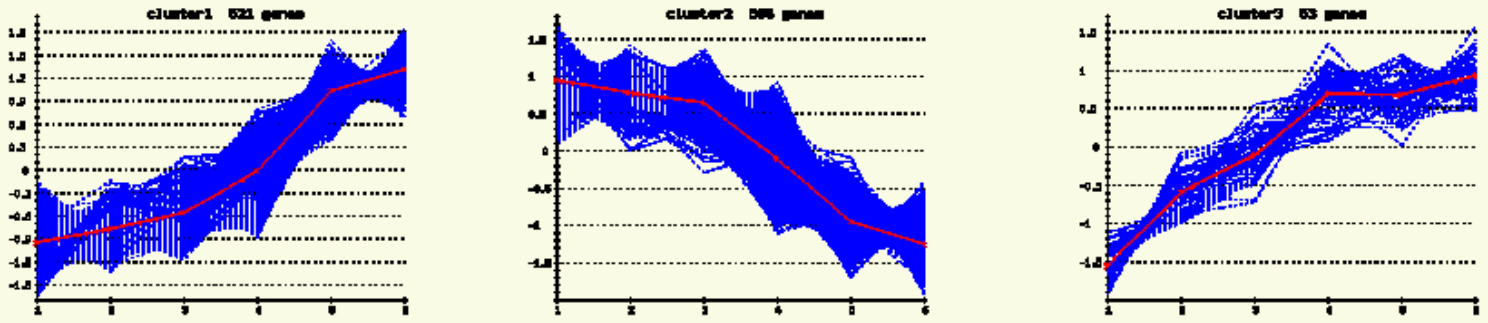
Applications of Clustering

- Data Mining
  - Technology watch
    - Derwent Database, contains all patents filed in the last 10 years worldwide
    - Searching by keywords leads to thousands of documents
    - Find clusters in the database and find if there are any emerging technologies and what competition is up to
  - Marketing
    - Customer database
    - Find clusters of customers and tailor marketing schemes to them

Application of Clustering

- Gene expression profile clustering
  - similar expressions , expect similar function

U18675 4CL -0.151 -0.207 0.126 0.359 0.208 0.091 -0.083 -0.209
M84697 a-TUB 0.188 0.030 0.111 0.094 -0.009 -0.173 -0.119 -0.136
M95595 ACC2 0.000 0.041 0.000 0.000 0.000 0.000 0.000 0.000
X66719 ACO1 0.058 0.155 0.082 0.284 0.240 0.065 -0.159 -0.010
U41998 ACT 0.096 -0.019 0.070 0.137 0.089 0.038 0.096 -0.070
AF057044 ACX1 0.268 0.403 0.679 0.785 0.565 0.260 0.203 0.252
AF057043 ACX2 0.415 0.000 -0.053 0.114 0.296 0.242 0.090 0.230
U40856 AIG1 0.096 -0.106 -0.027 -0.026 -0.005 -0.052 0.054 0.006
U40857 AIG2 0.311 0.140 0.257 0.261 0.158 0.056 -0.049 0.058
AF123253 AIM1 -0.040 0.002 -0.202 -0.040 0.077 0.081 0.088 0.224
X92510 AOS 0.473 0.560 0.914 0.625 0.375 0.387 0.019 0.141



From:De Smet F., Mathys J., Marchal K., Thijs G., De Moor B. & Moreau Y. 2002. *Adaptive Quality-based clustering of gene expression profiles*, Bioinformatics, **18**(6), 735-746.

Applications of Clustering

- Profiling Web Users
  - Use web access logs to generate a feature vector for each user
  - Cluster users based on their feature vectors
  - Identify common goals for users
    - Shopping
    - Job Seekers
    - Product Seekers
    - Tutorials Seekers
  - Can use clustering results to improving web content and design

Distance and Similarity measures in Clustering

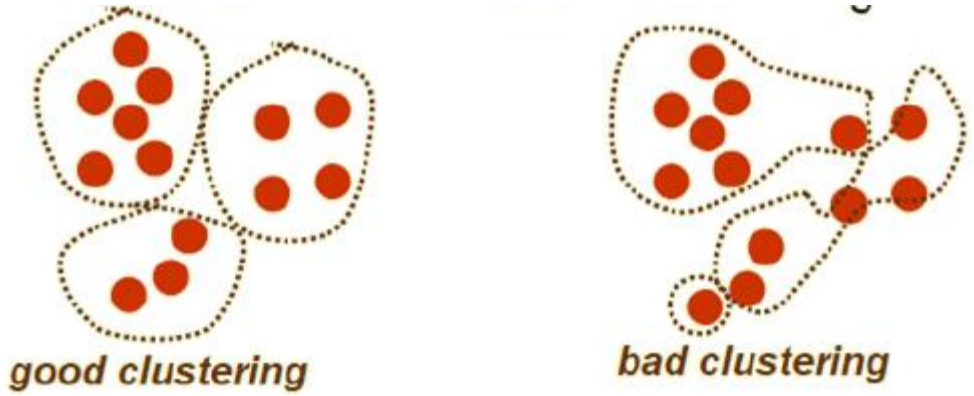
Distance Measures:

- Distance measures are used to determine the dissimilarity between two data points.
- There are several distance measures commonly used in clustering.

1. Minkowski distance family
    1. Euclidean Distance
    2. Manhattan Distance
3. Cosine Similarity
4. Mahalanobis Distance

What do we need for clustering?

1. Proximity measure,
   - **Similarity measure $s(x_i, x_j)$**: if x_i, x_j are similar - Higher the similarity
   - **Dissimilarity (distance) measure $d(x_i, x_j)$**: if x_i, x_j are similar – difference is small
2. Criterion function to evaluate a clustering
   - Clustering – good or bad?
3. Algorithm to compute clustering
   - Optimizing the criterion function



Distance (dissimilarity) measures

- They are special cases of **Minkowski distance**:

$$d_p(\mathbf{x}_i, \mathbf{x}_j) = \left(\sum_{k=1}^{m} \left| x_{ik} - x_{jk} \right|^p \right)^{\frac{1}{p}}$$

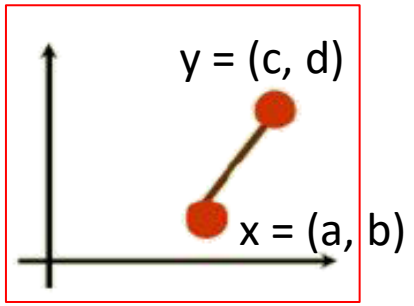
(p is a positive integer)

Dissimilarity (Distance) Measures

Euclidean: Take the square root of the sum of the squares of the differences of the coordinates

For example, if x=(a, b) and y=(c, d), the Euclidean distance between x and y is

$$d = \sqrt{(a-c)^2 + (b-d)^2}$$



In general, the Euclidean distance d (x_i, y_i) is

$$d\,(x_i, y_i) = \sqrt{\sum_{k=1}^{d} (x_i - y_i)^2}$$

Distance (dissimilarity) measures

Manhattan: Take the sum of the absolute values of the differences of the coordinates. The Manhattan distance is the sum of the horizontal and vertical components.

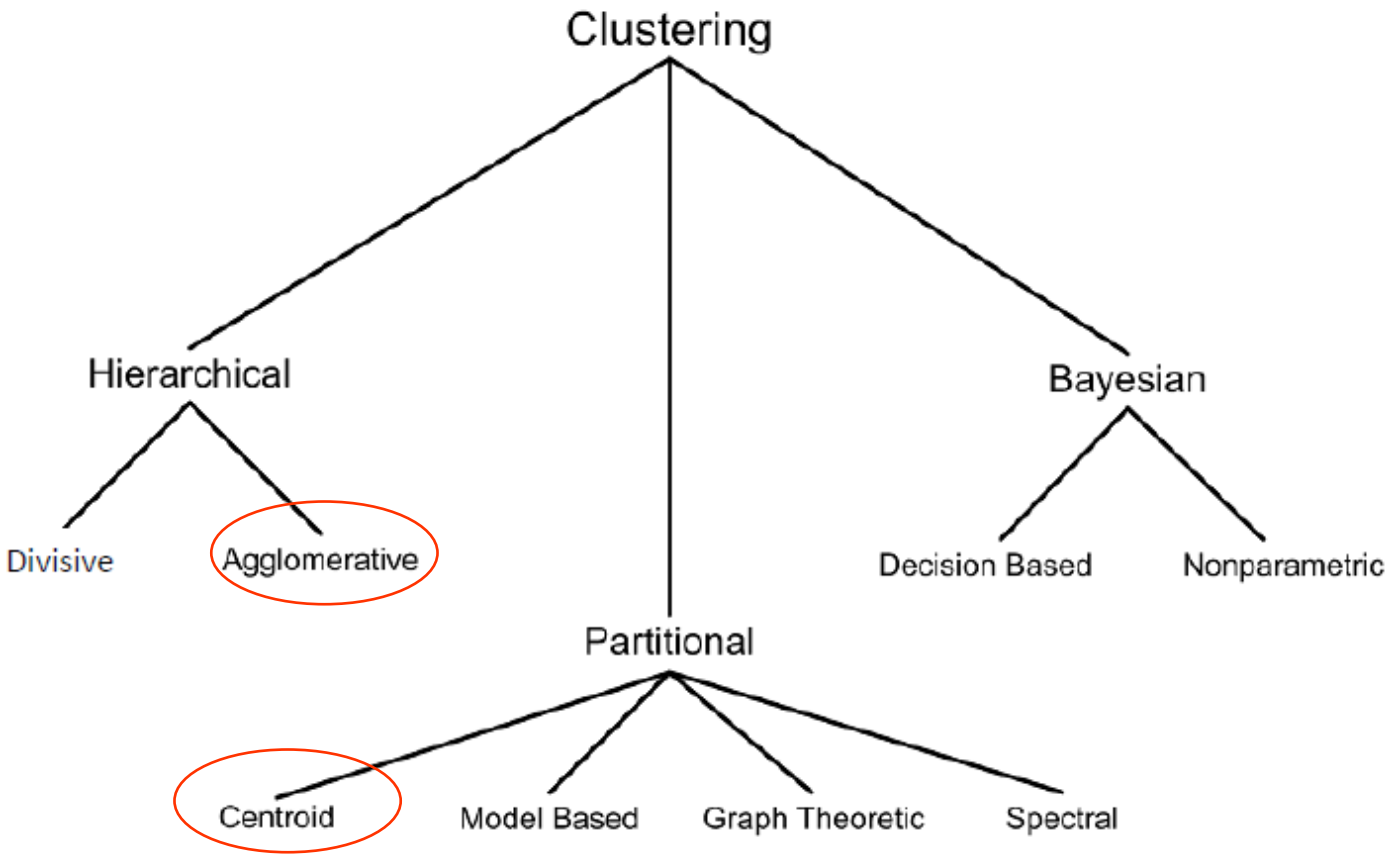
For example, if x=(a, b) and y=(c, d), the Manhattan distance between x and y is

$$d = |a - c| + |b - d|$$

In general, the Manhattan distance d (x_i, x_j) is

$$d\,(x_i, x_j) = \sum_{k=1}^{d} \left(|x_i^{\,k} - x_j^{\,k}| \right)$$

Clustering Techniques



Clustering Techniques

- Hierarchical algorithms find successive clusters using previously established clusters.
- Types:
  - Agglomerative (“bottom-up”) or
  - Divisive (“top-down”)

  - Agglomerative algorithm begins with each element as a separate cluster and merge them into successively larger clusters

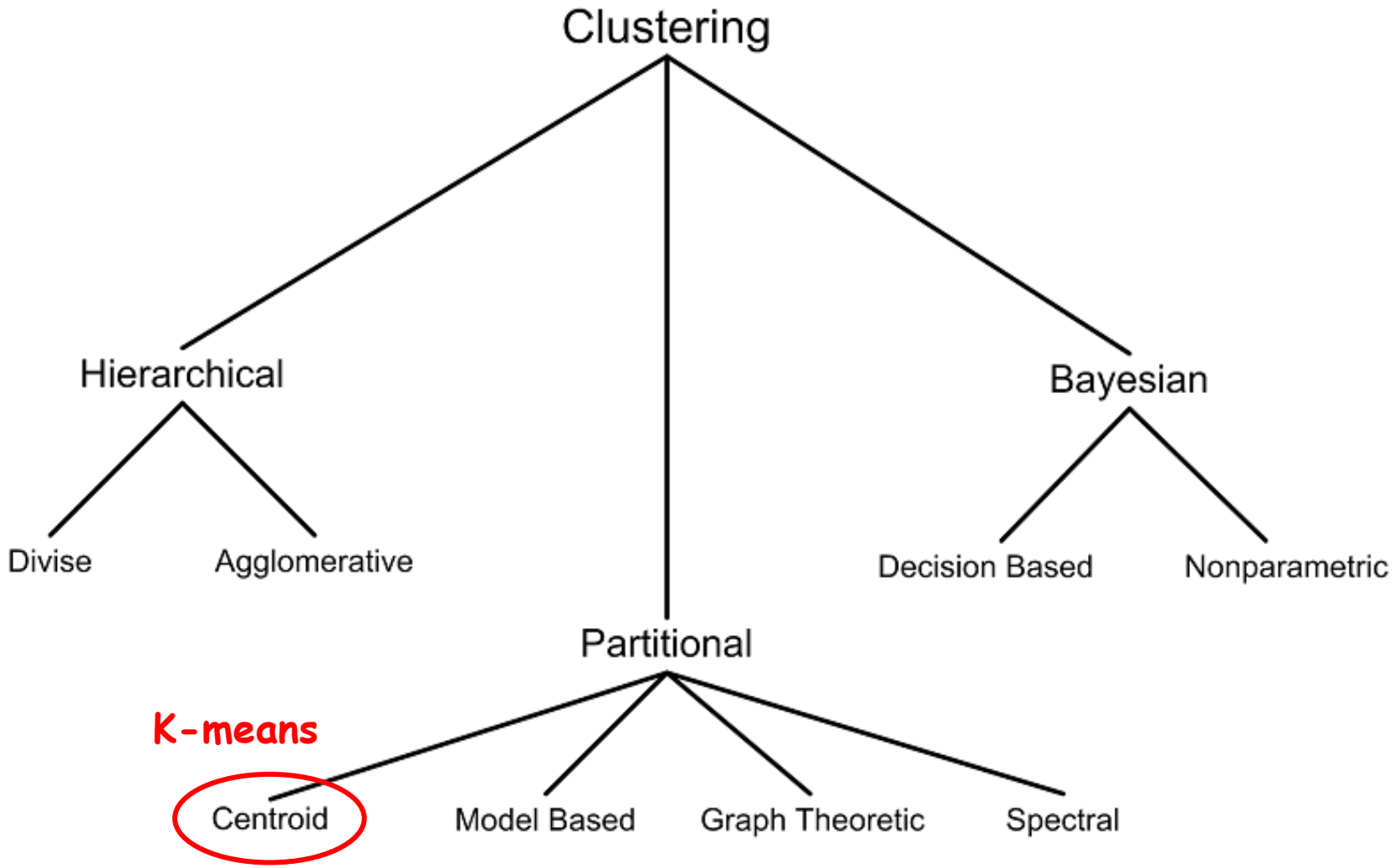
  - Divisive algorithms begin with the whole set proceed to divide it into successively smaller clusters.

Clustering Techniques

- Partitional algorithm typically determine all clusters at once, but can also be used as divisive algorithms in the hierarchical clustering.

- Bayesian algorithm try to generate a posteriori distribution over the collection of partitions of the data.

Clustering Techniques



K-Means Clustering

MacQueen, 1967

K-means is a Partitional Clustering algorithm

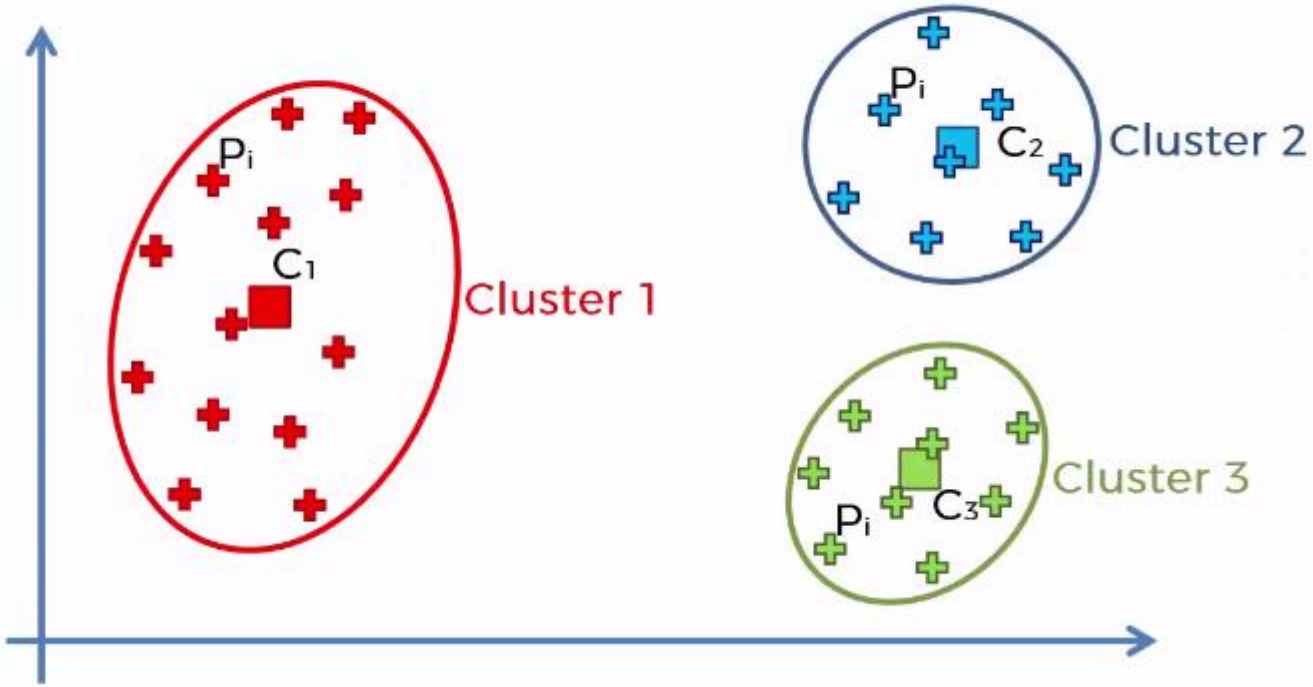
The *k*-means algorithm partitions the given data into *k* clusters:

- Each cluster has a cluster **center**, called **centroid**.
- *k* is specified by the user (K is a positive integer)

The grouping is done by minimizing the sum of squares of distances between data and the corresponding cluster centroid.

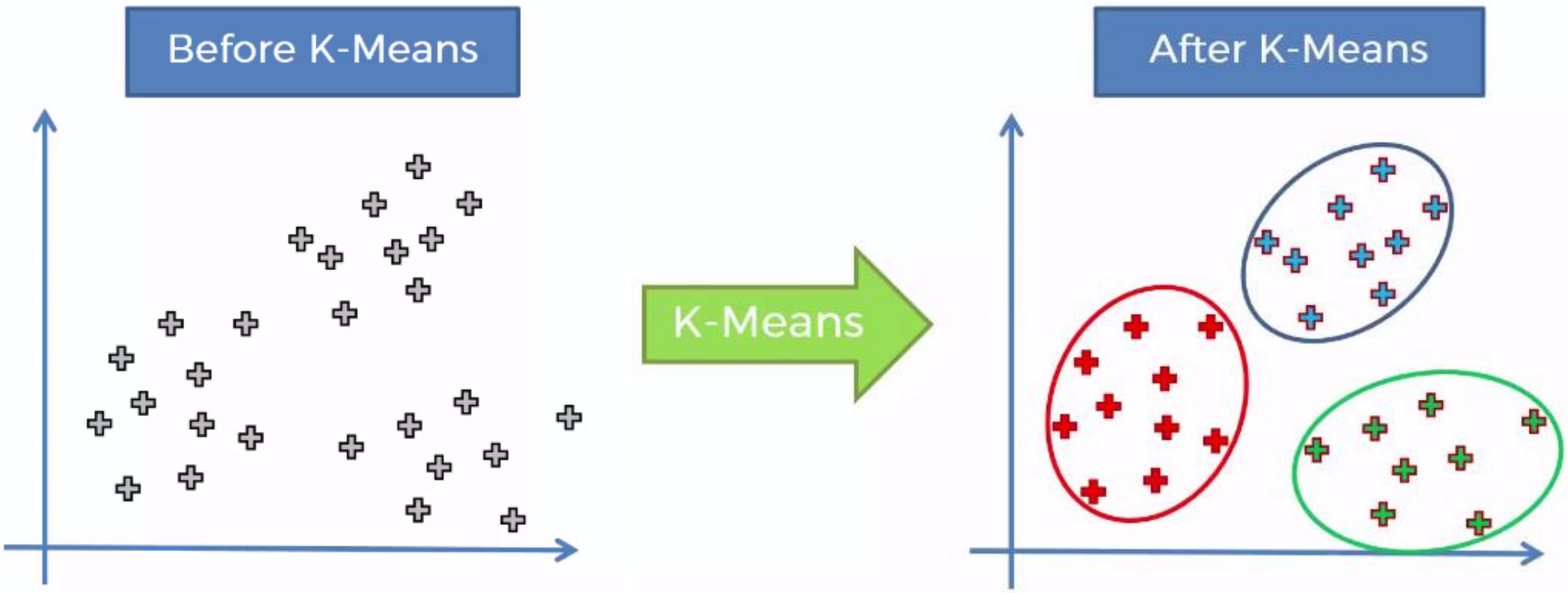
K-means clustering

- Given k,
  - Objects $X = \{x_1, x_2, \ldots, X_m\}$
  - Each object $x_i = \{x_{i1}, x_{i2}, \ldots, X_{in}\}$
  - Output clusters $S = = \{S_1, S_2, \ldots, S_k\}$
  - S_i is represented by cluster centre μ_i optimizes using some chosen criteria (e.g. WCSS should be minimum)



$$\text{WCSS} = \sum_{p_i \text{ in cluster 1}} \text{distance}\,(P_i, C1)^2 + \sum_{p_i \text{ in cluster 2}} \text{distance}\,(P_i, C_2)^2 + \sum_{p_i \text{ in cluster 3}} \text{distance}\,(P_i, C_3)^2$$

K-Means Clustering



K-Means Algorithm

Step-1: Choose the number of clusters (K)

Step-2: Select at random k points, the centroids (not necessarily from your dataset)

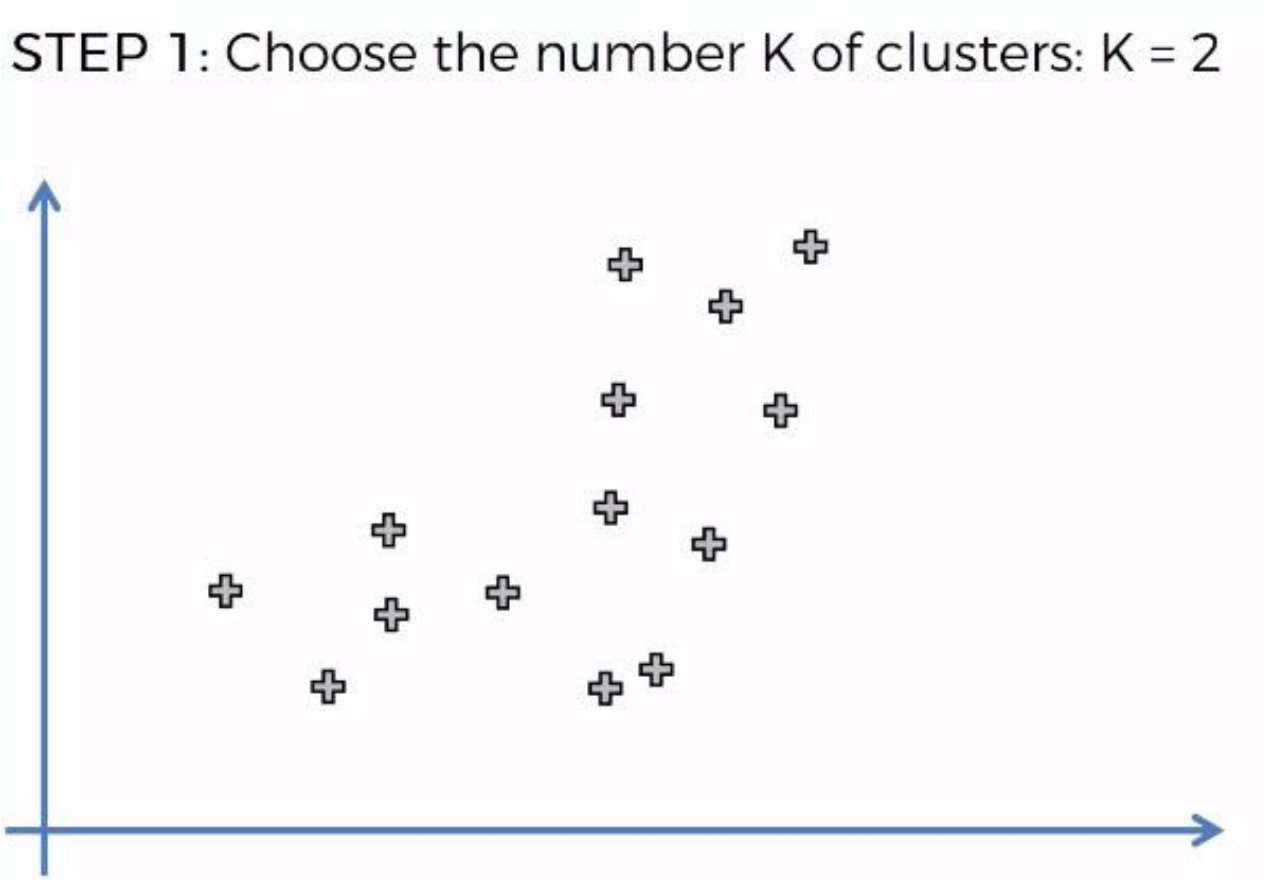
Step-3: Assign each data point to the closest centroid → that forms k clusters

Step-4: Compute and place the new centroid of each cluster

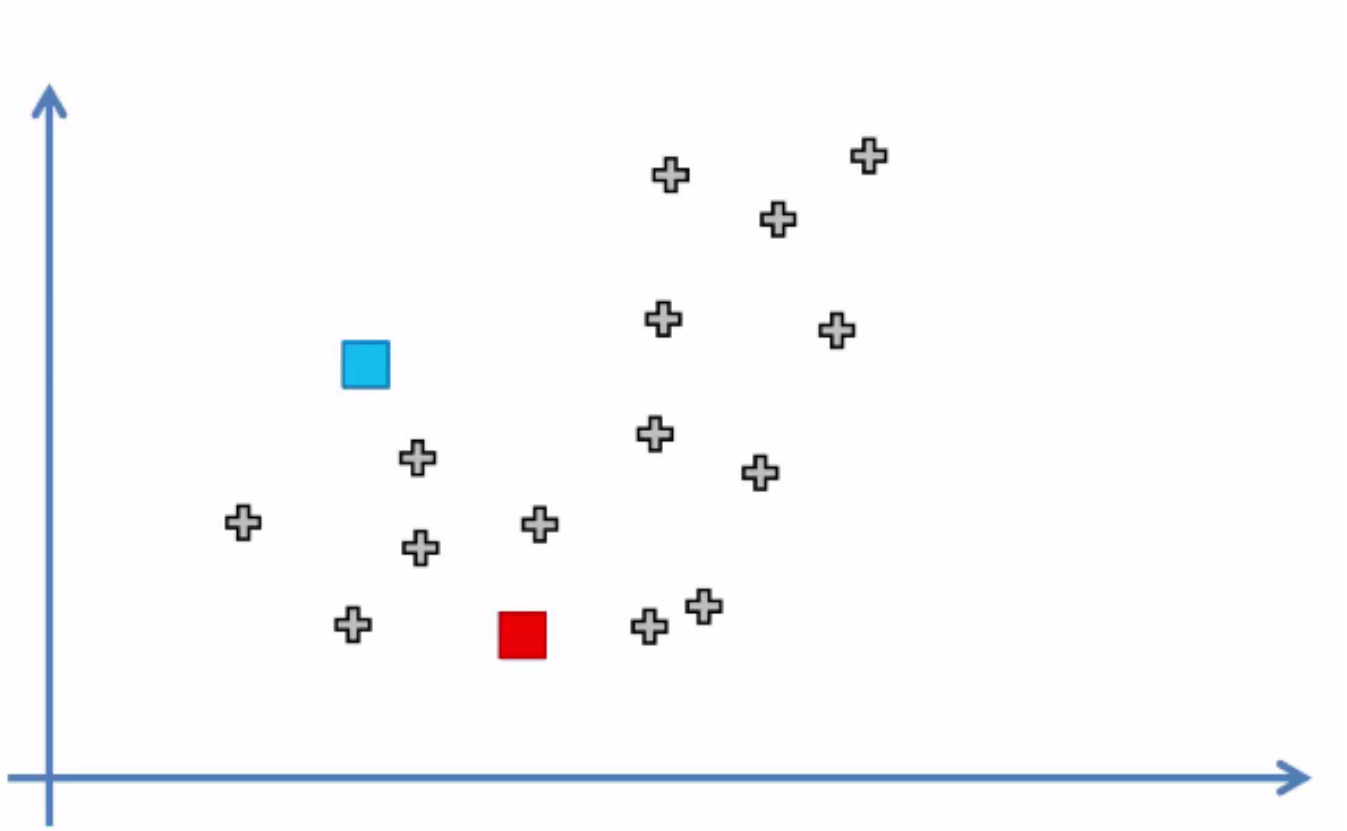
Step-5: Reassign each data points to the new closest centroid.
If any reassignment took place, go to Step-4, otherwise go to Stop.

Stop: your Model is ready

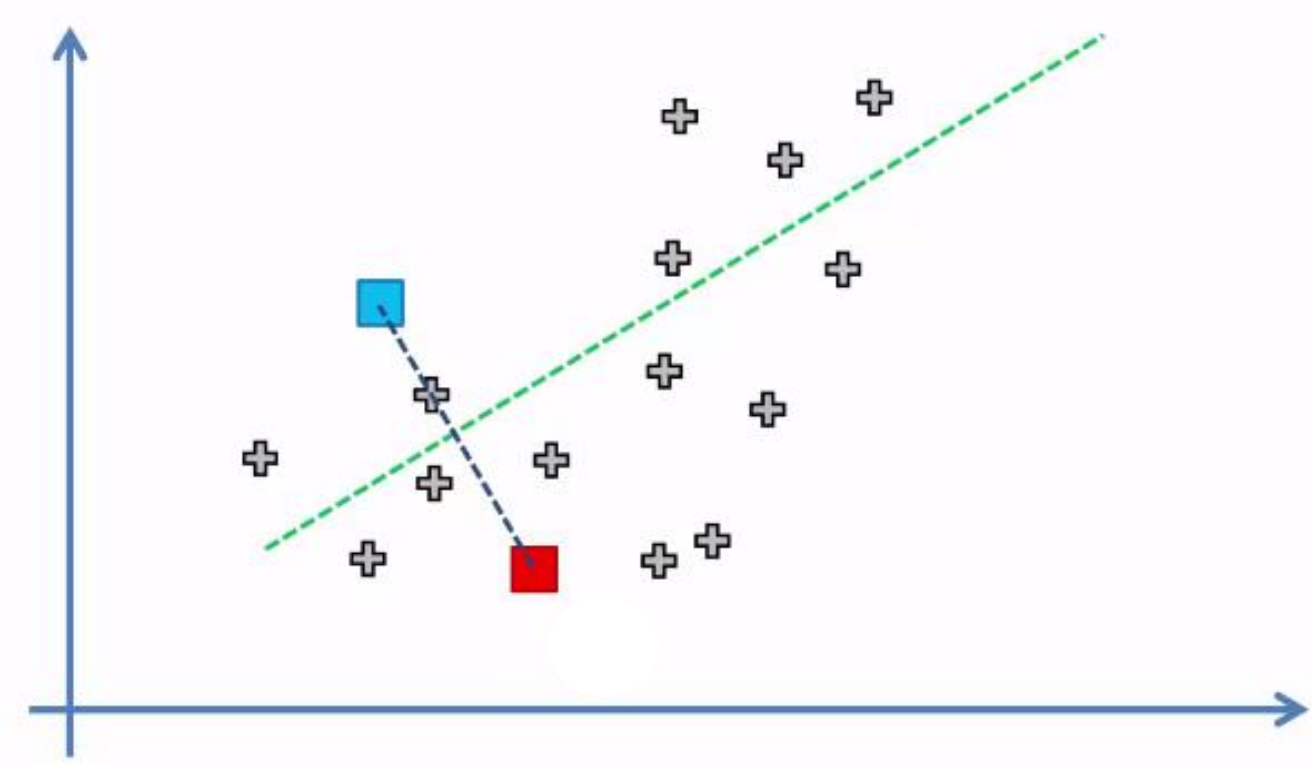
Example



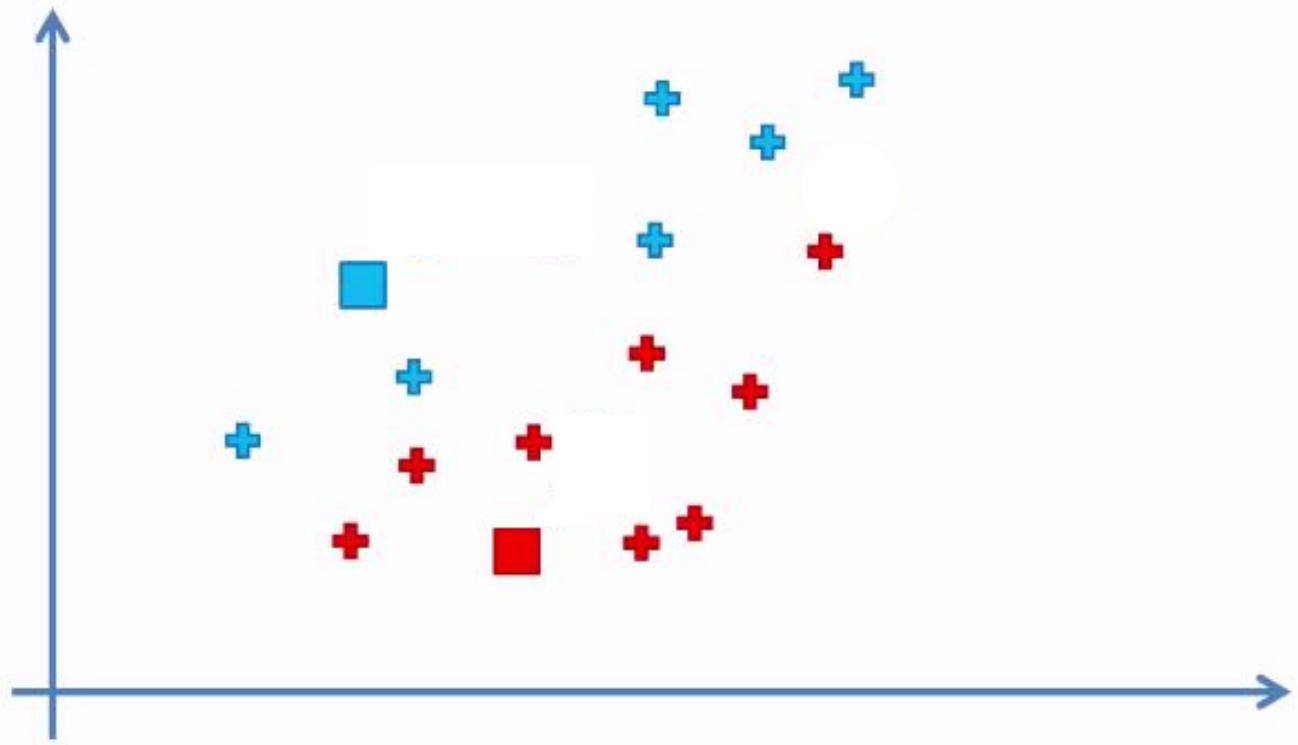
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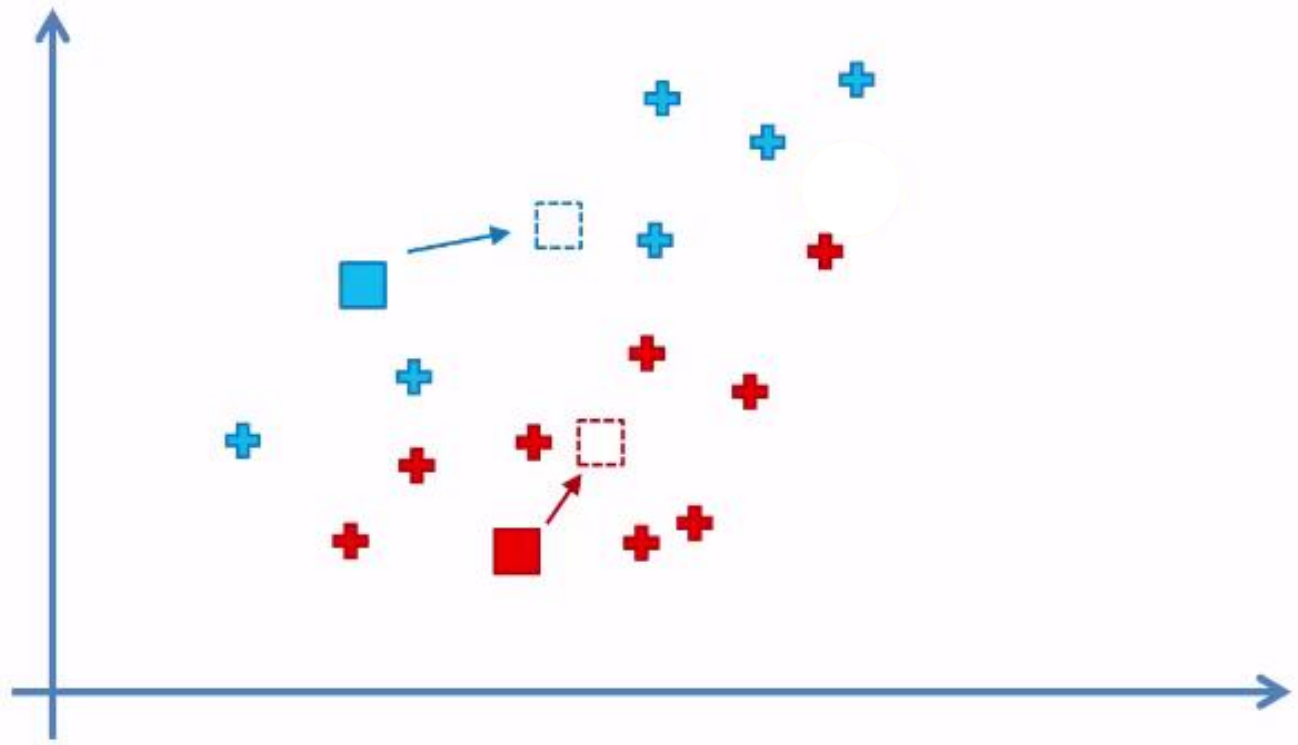
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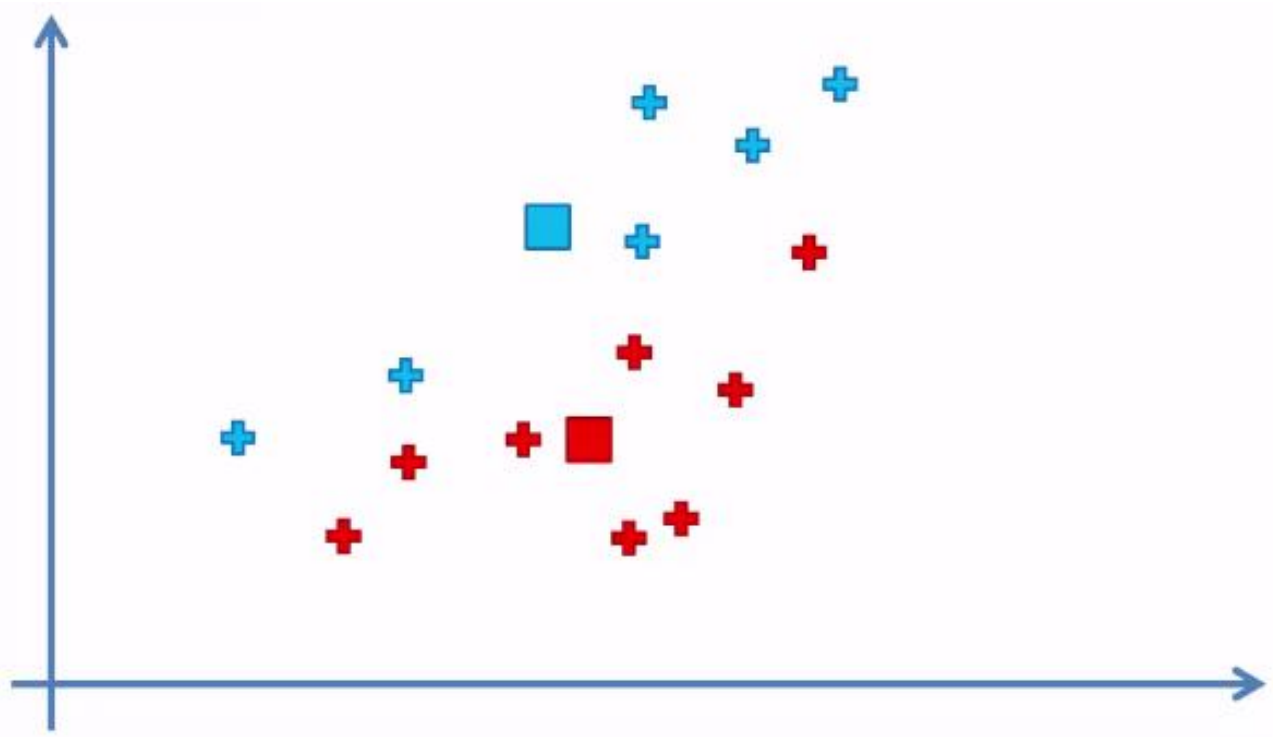
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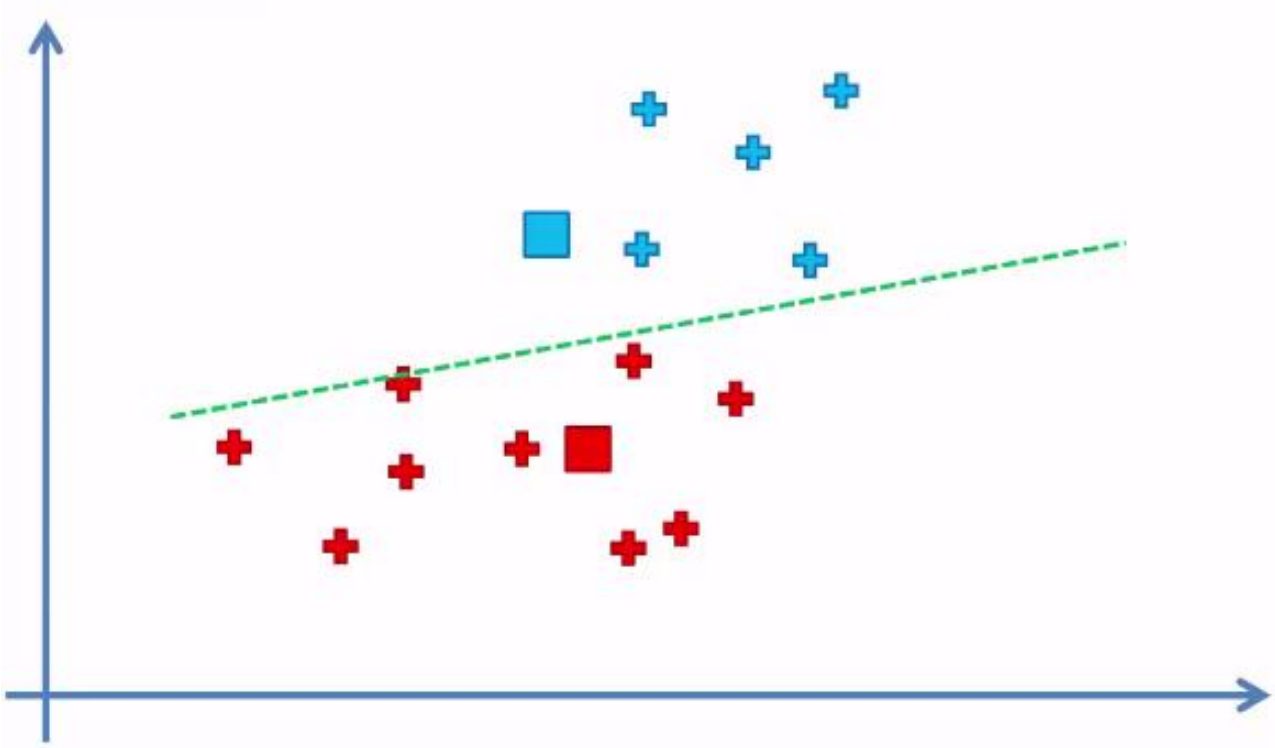
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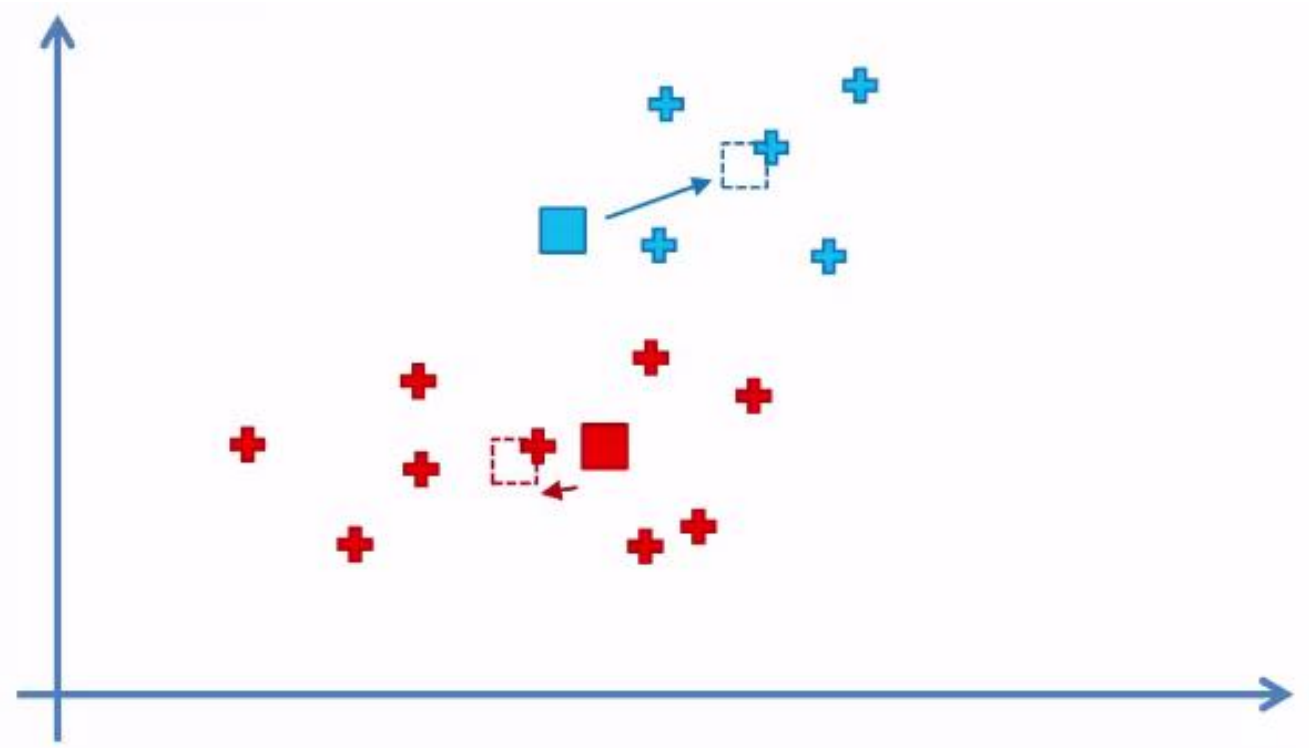
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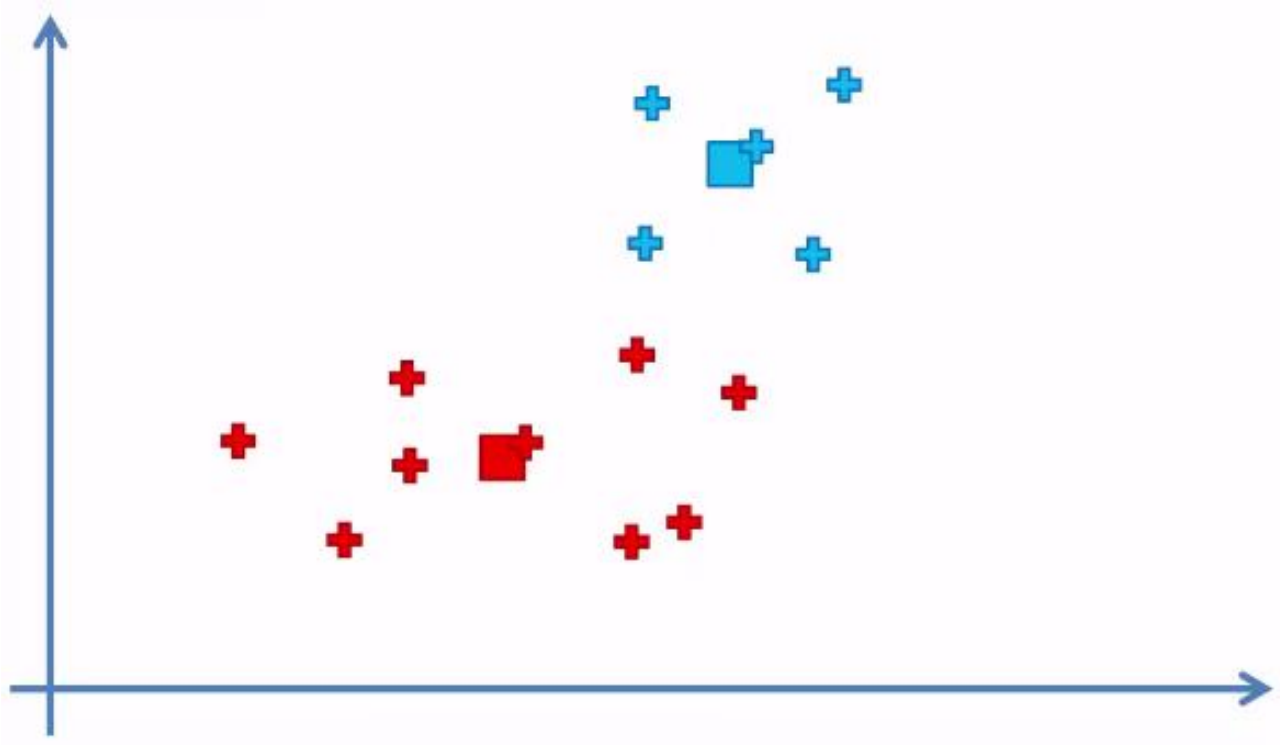
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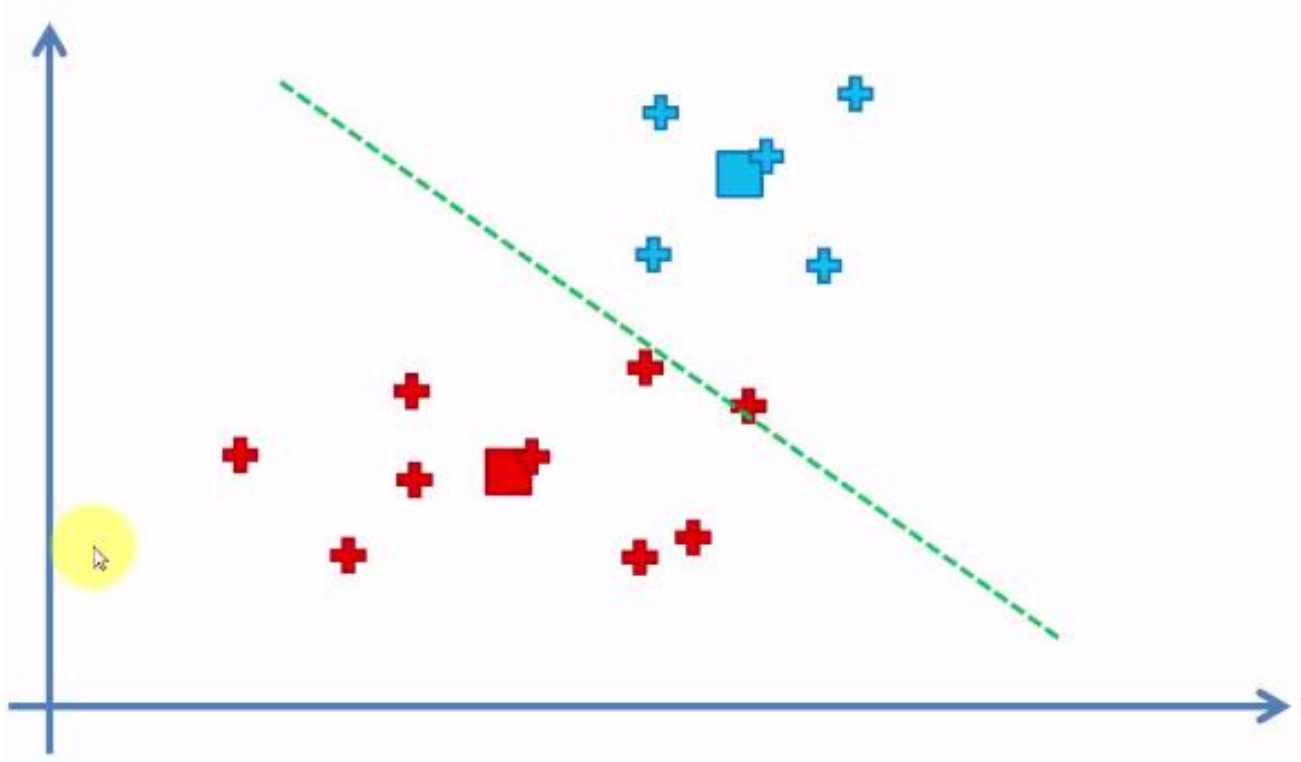
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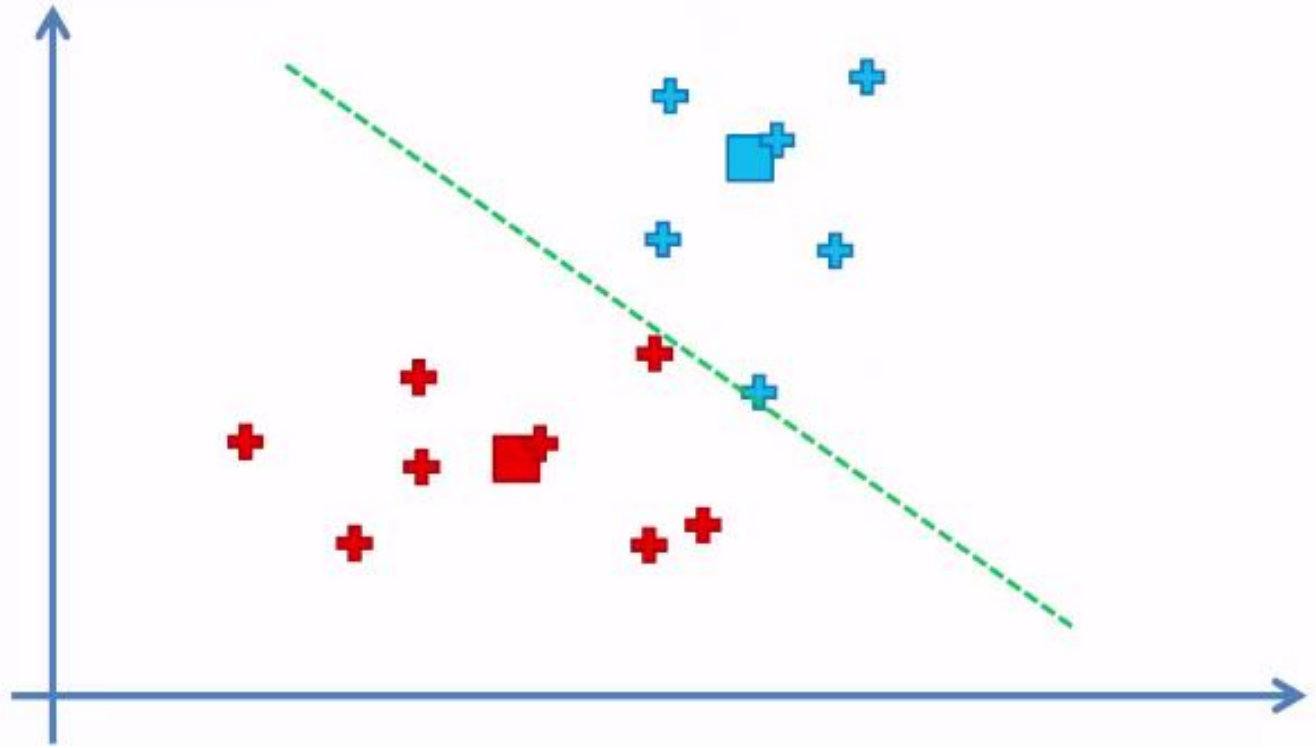
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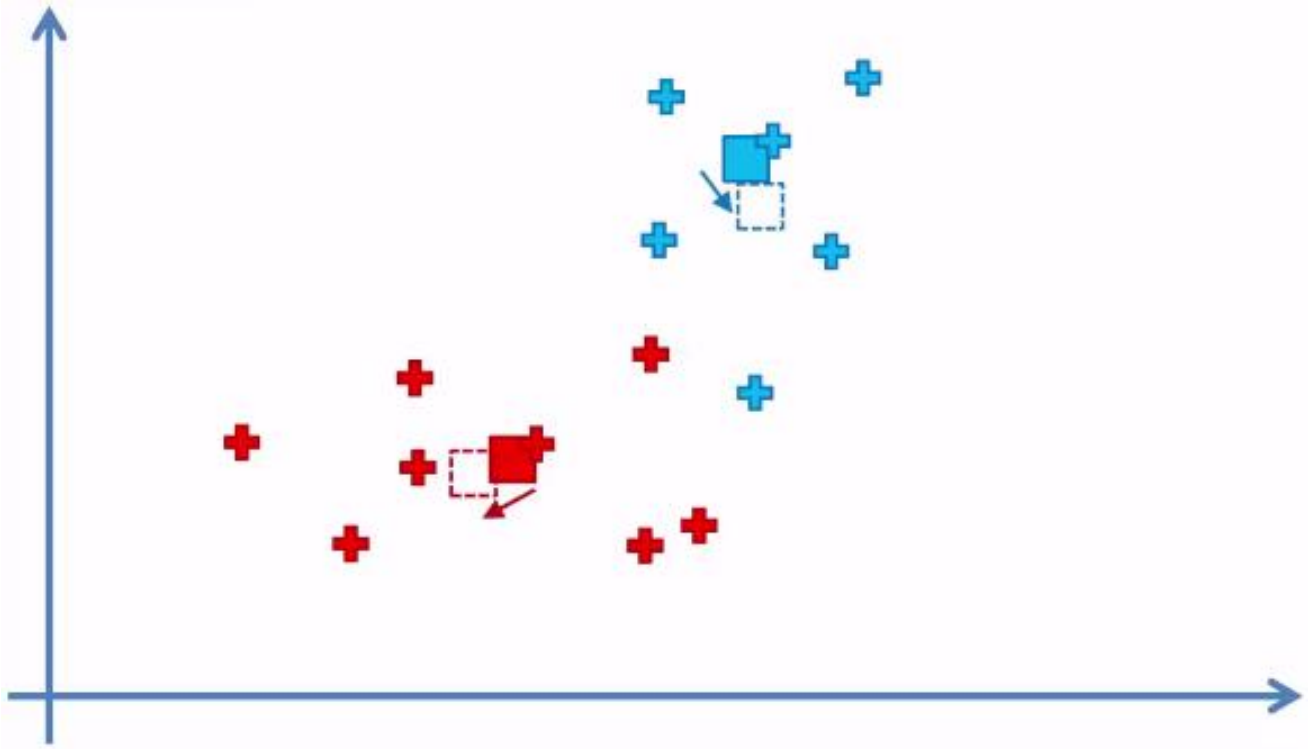
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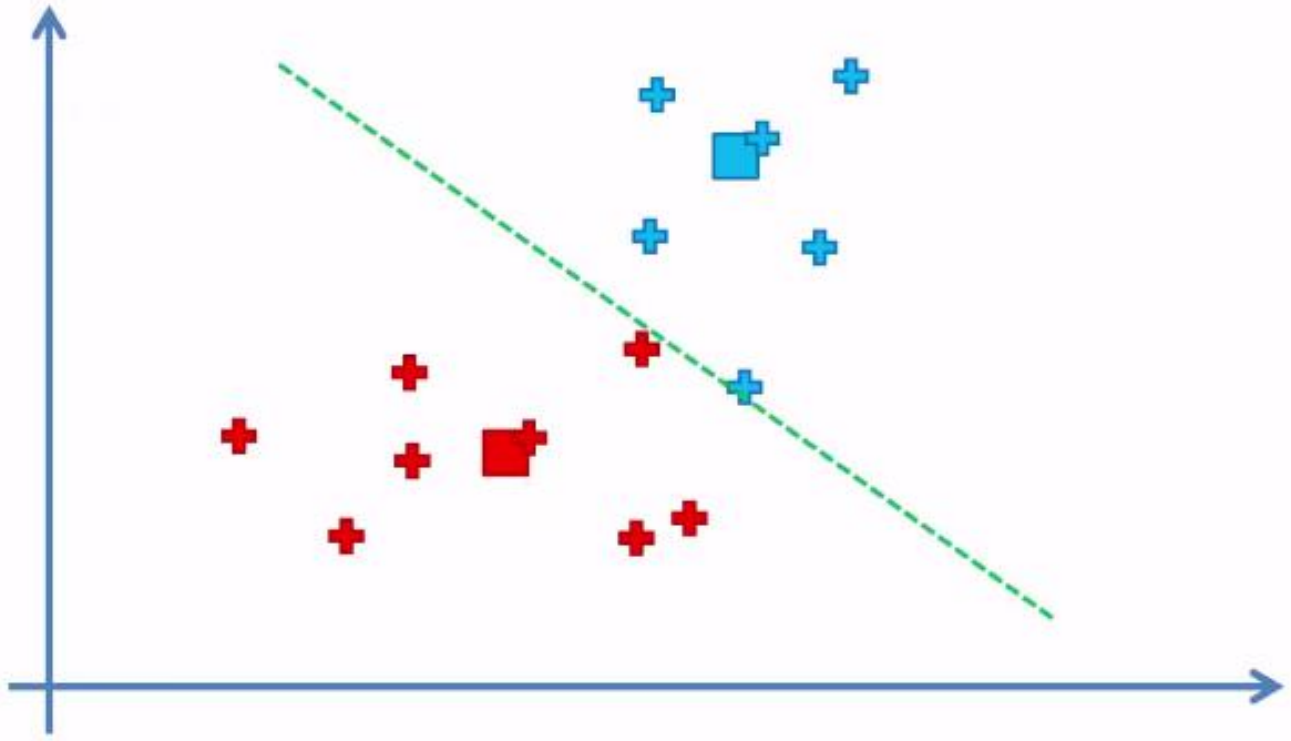
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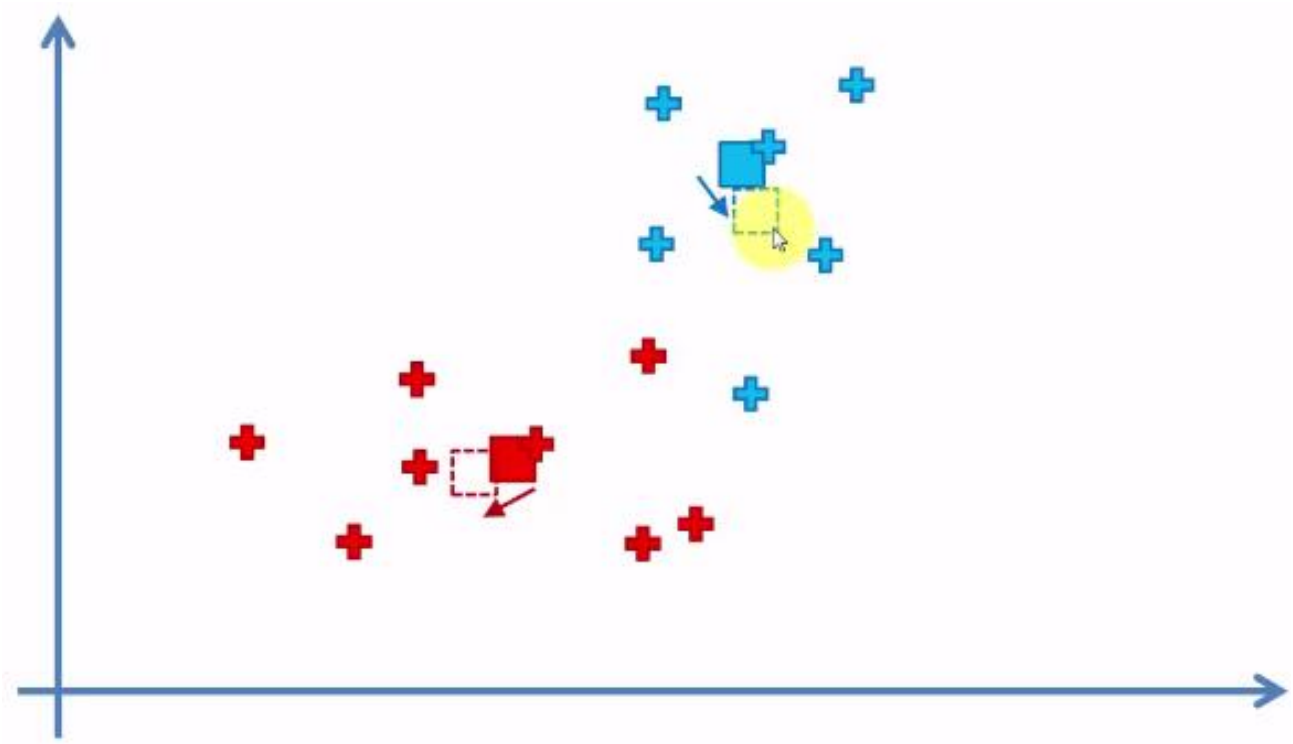
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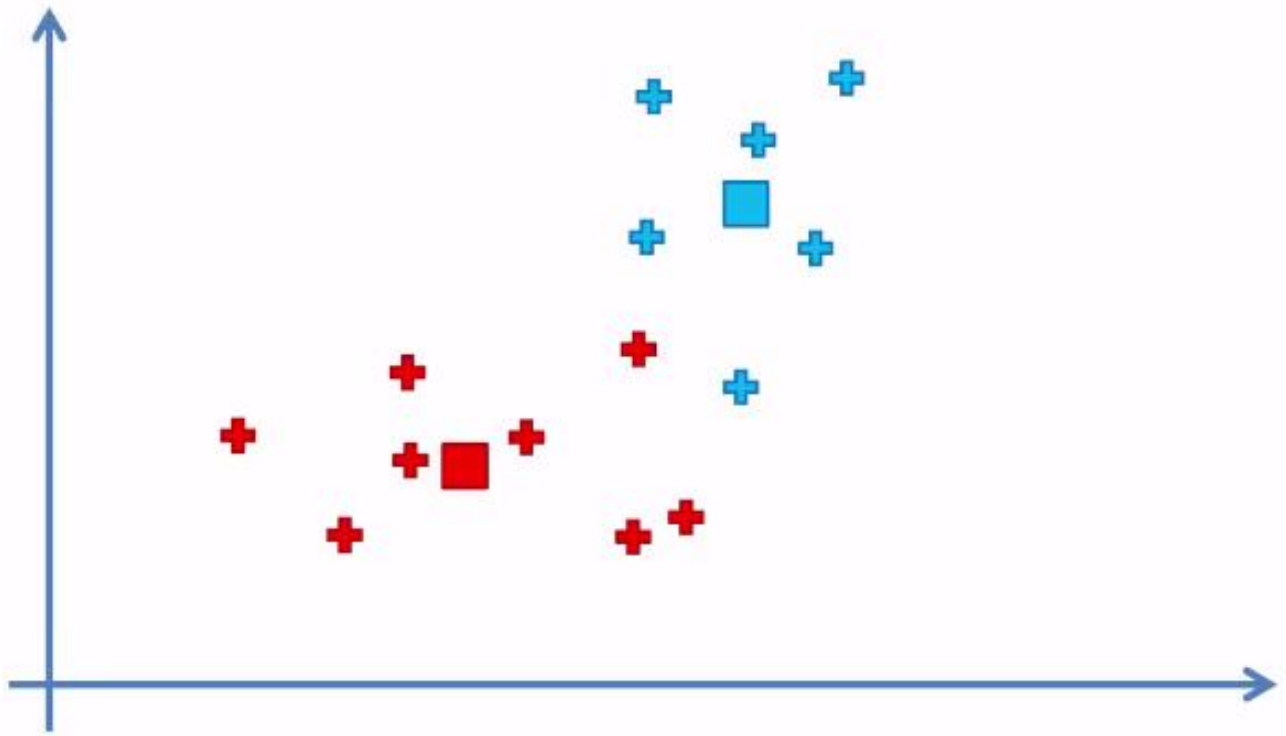
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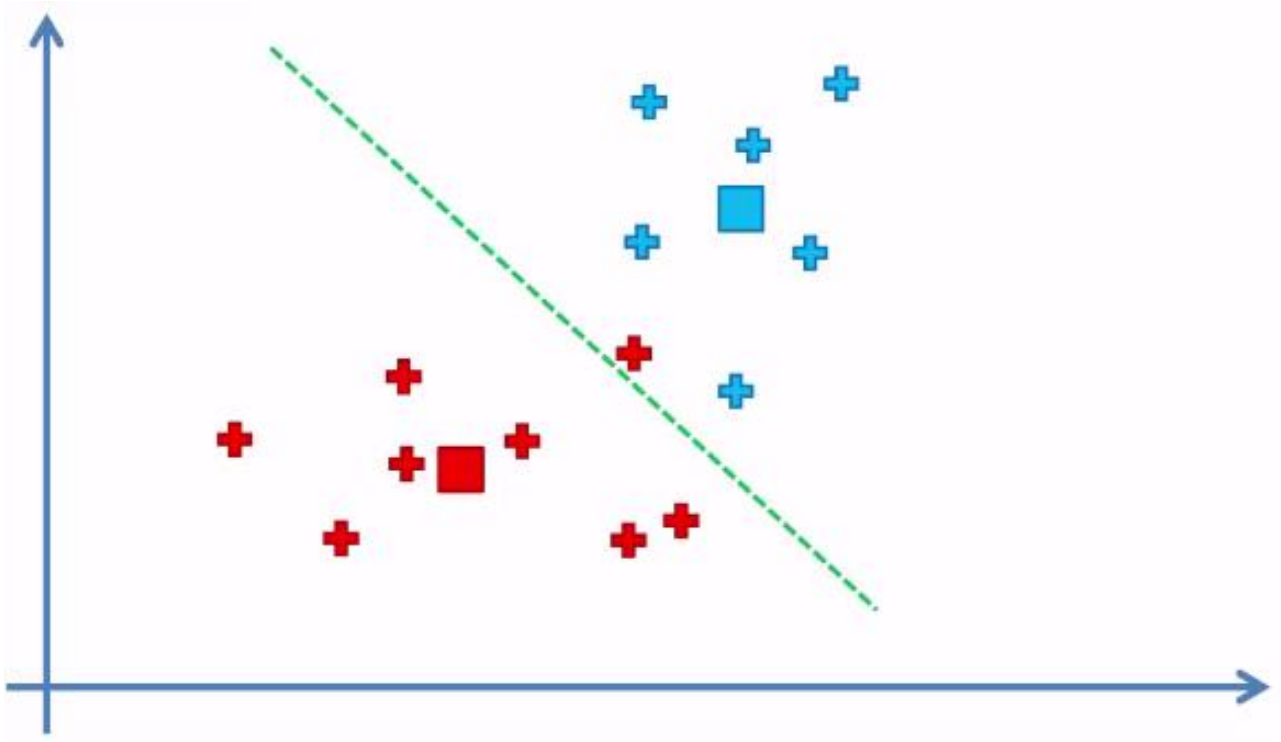
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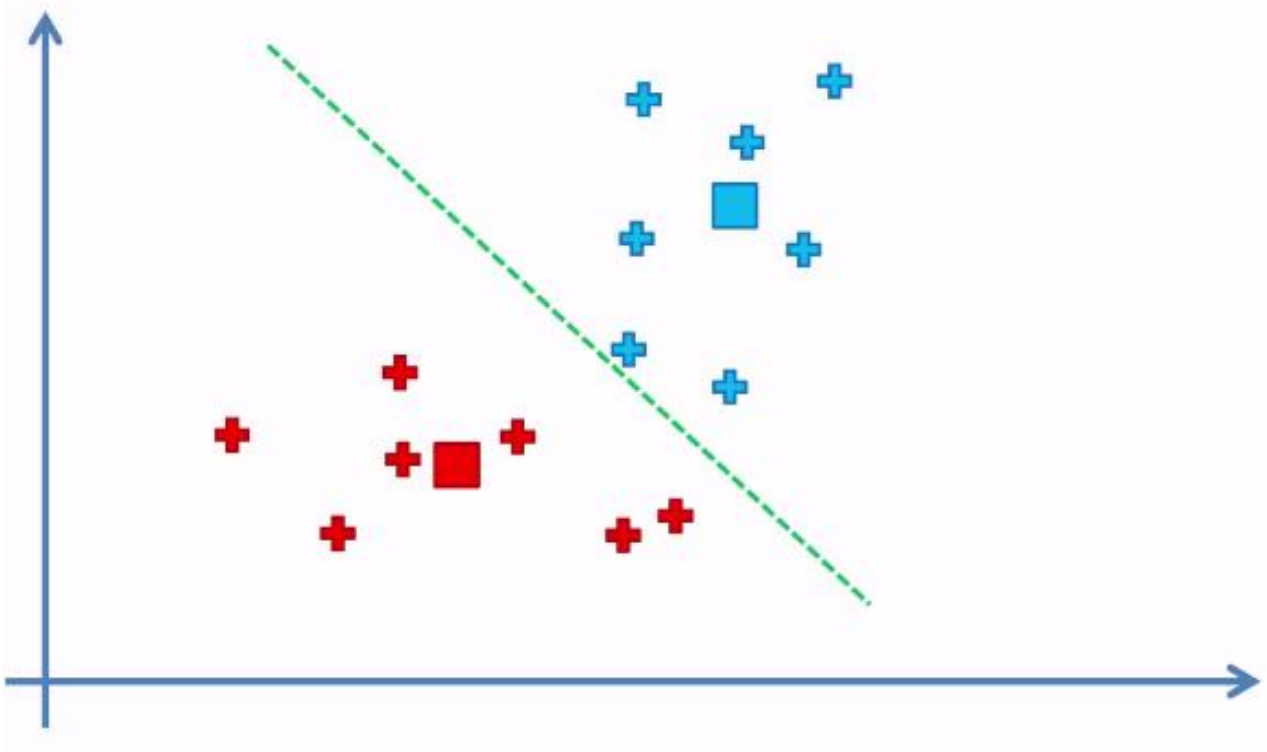
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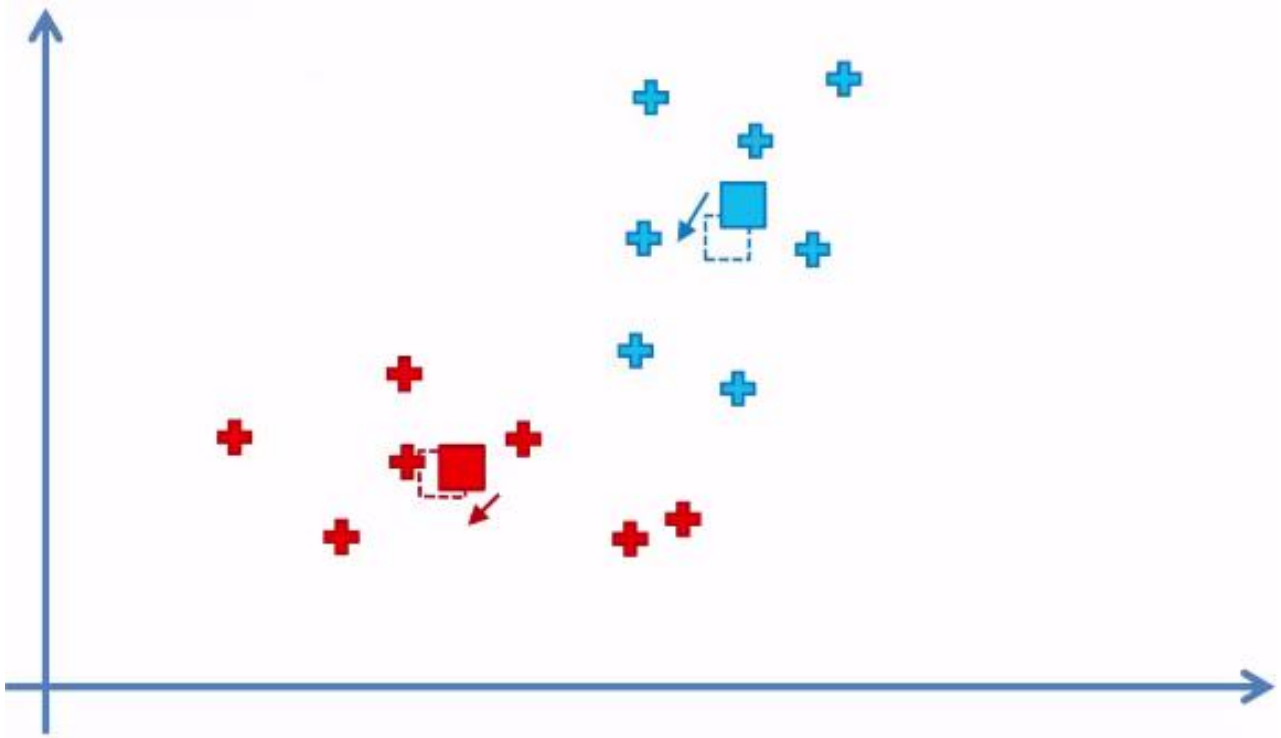
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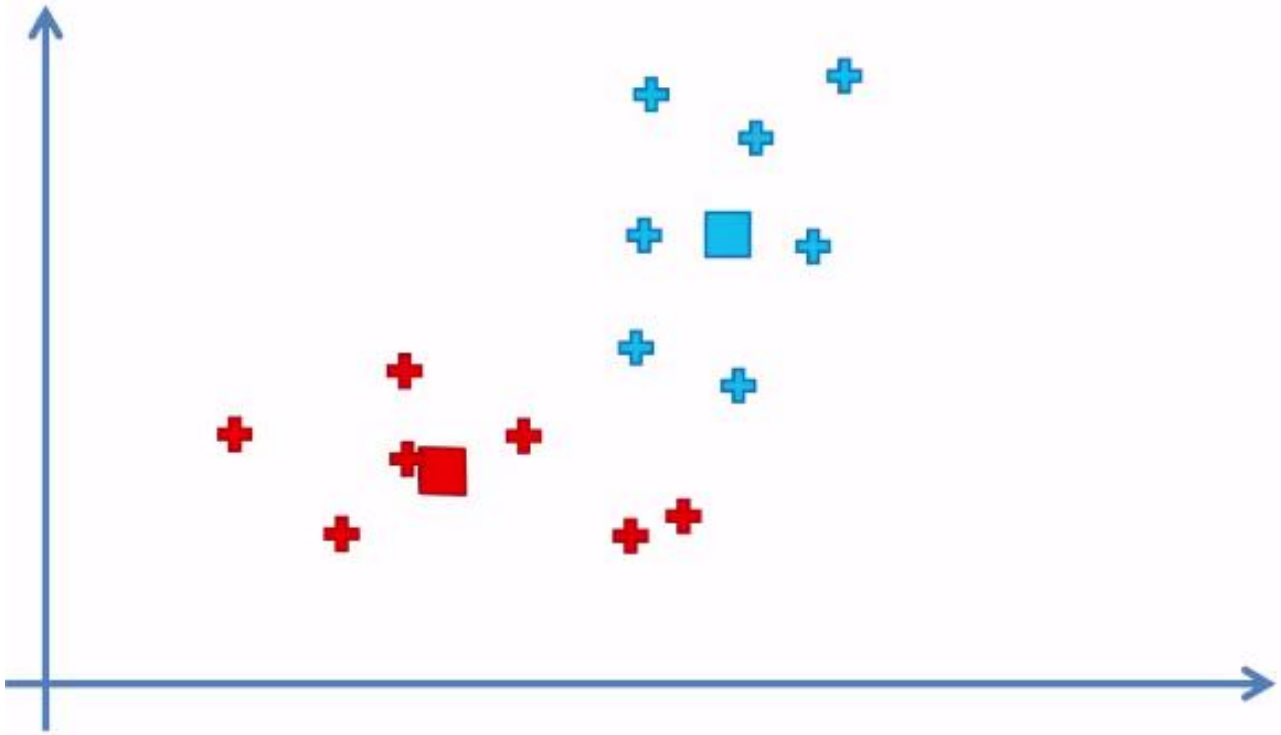
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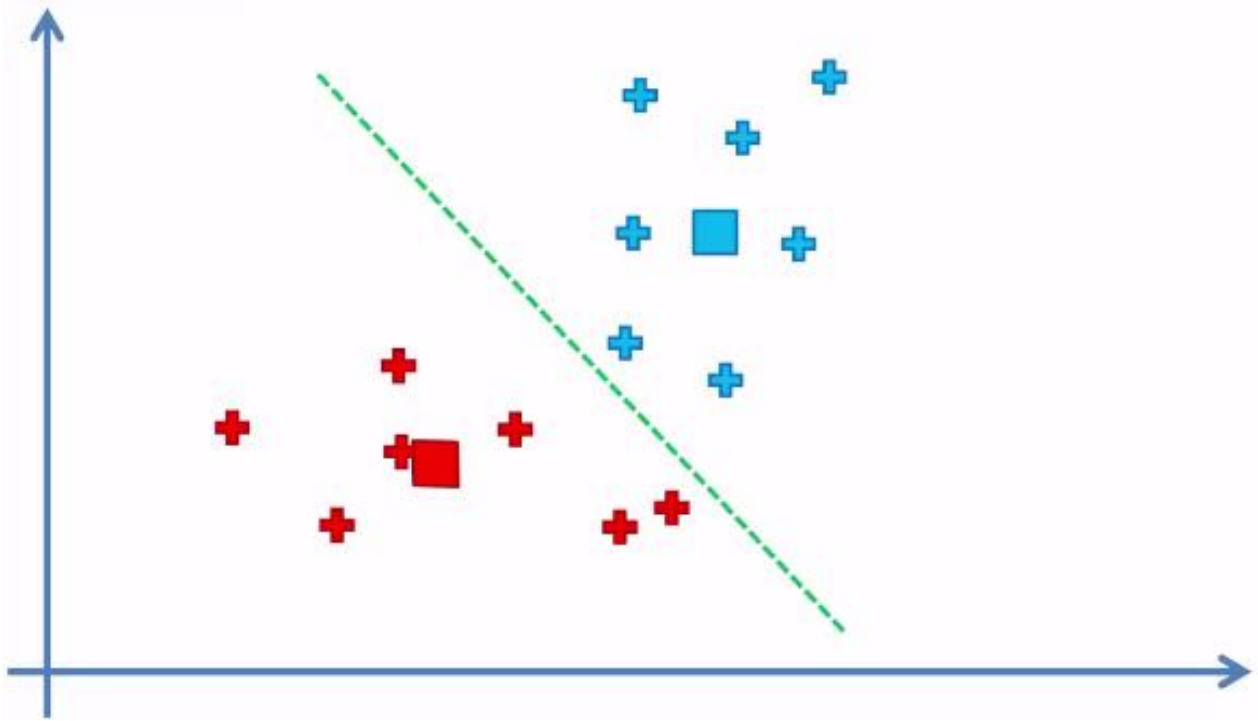
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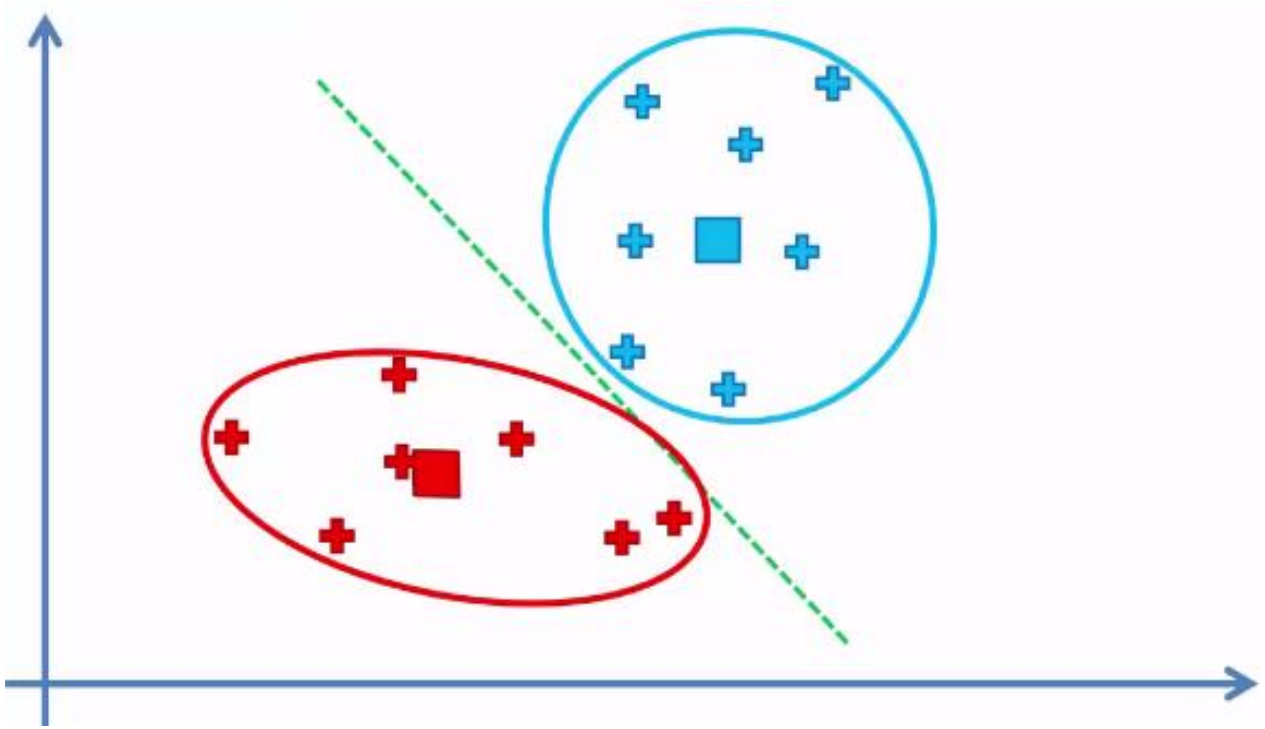
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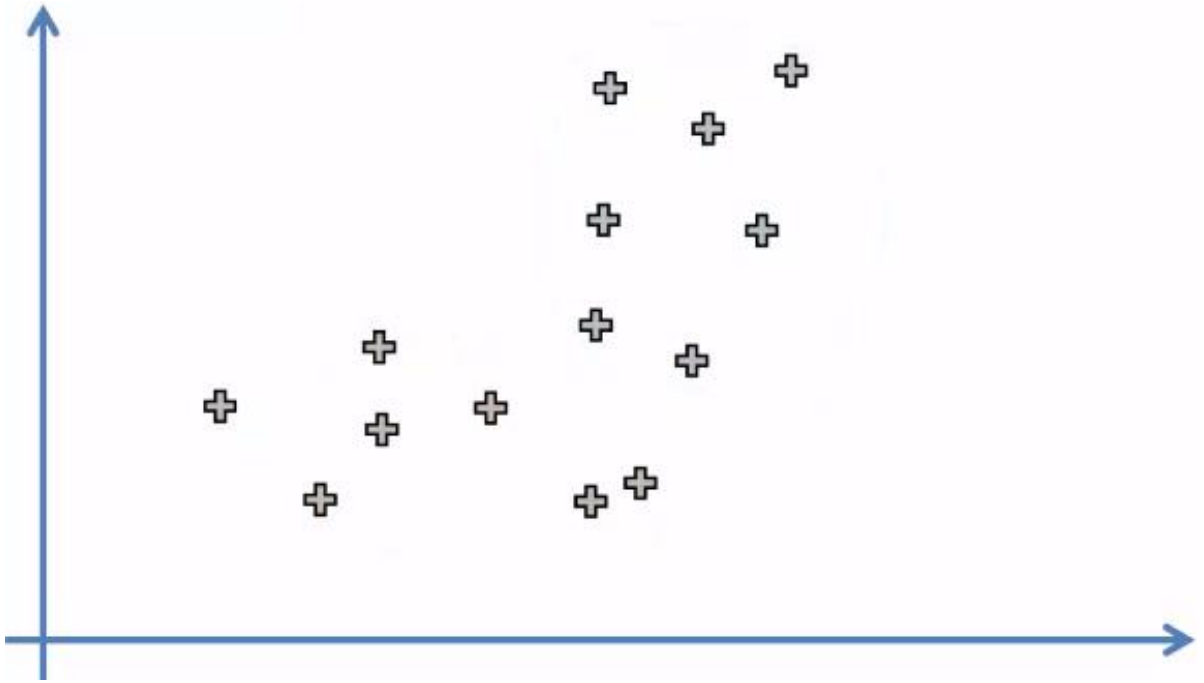
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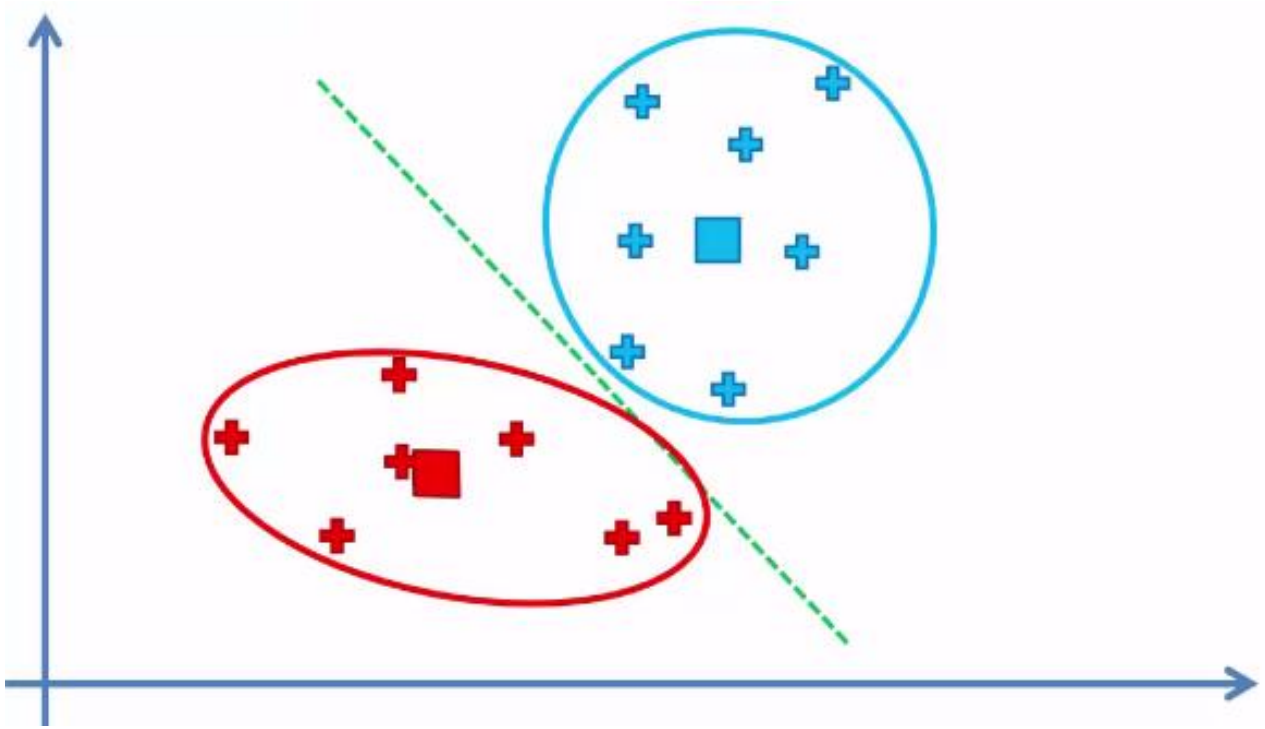
Stop: your model is ready



Step-2: Select at random k points, the centroids (not necessarily from your dataset)

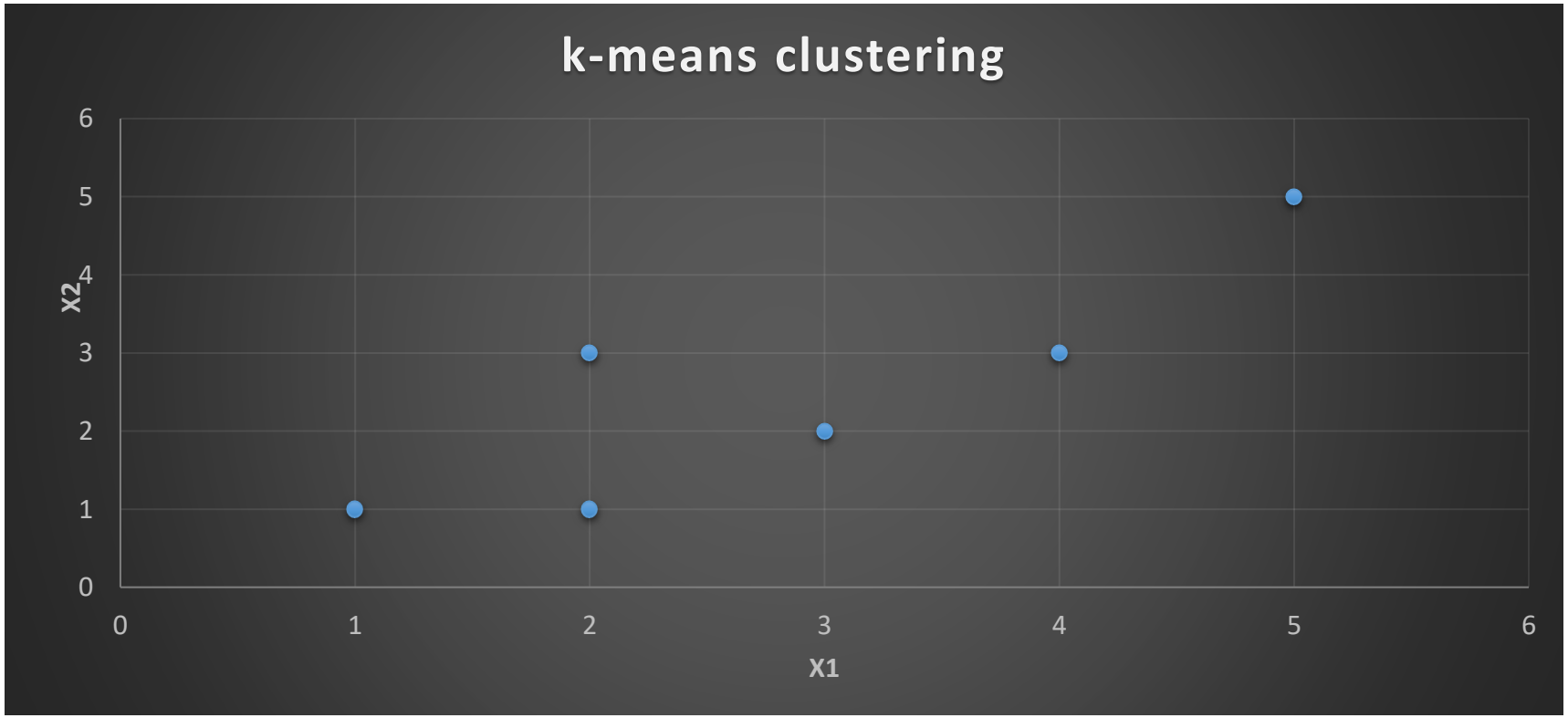


Stop: your model is ready



Example: Use k-means clustering algorithm to divide the following data points into 2 clusters

X1	1	2	2	3	4	5
X2	1	1	3	2	3	5

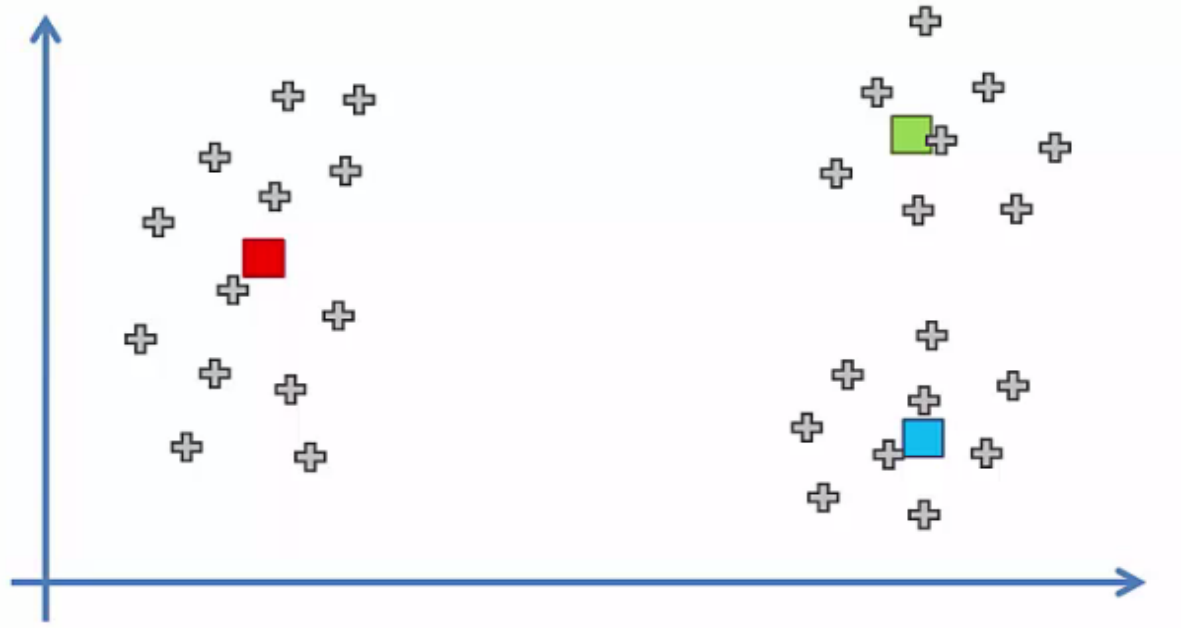


Random Initialization of Centroids

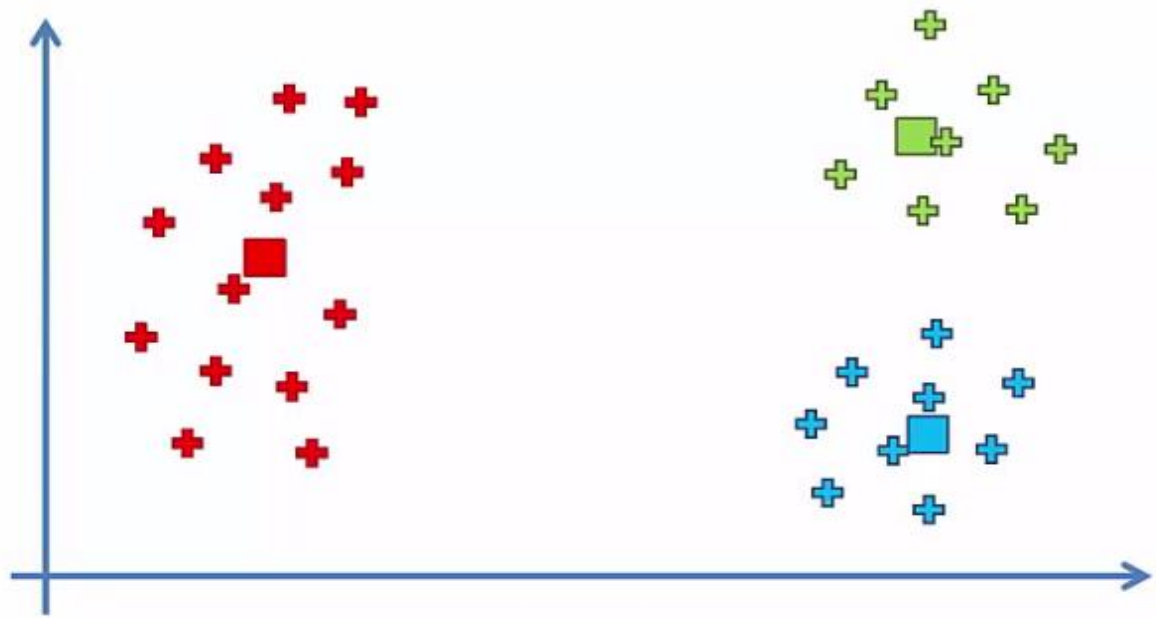


If we choose K = 3 clusters...

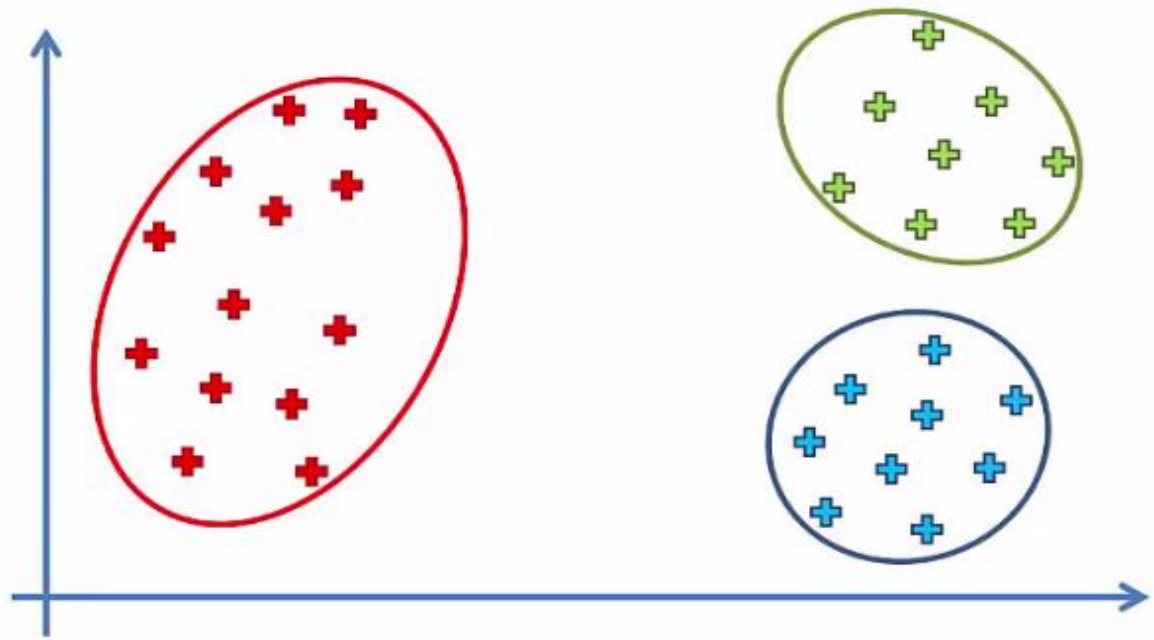
If I chose correct random initialization of centroids …



…this correct random initialisation would lead us to…



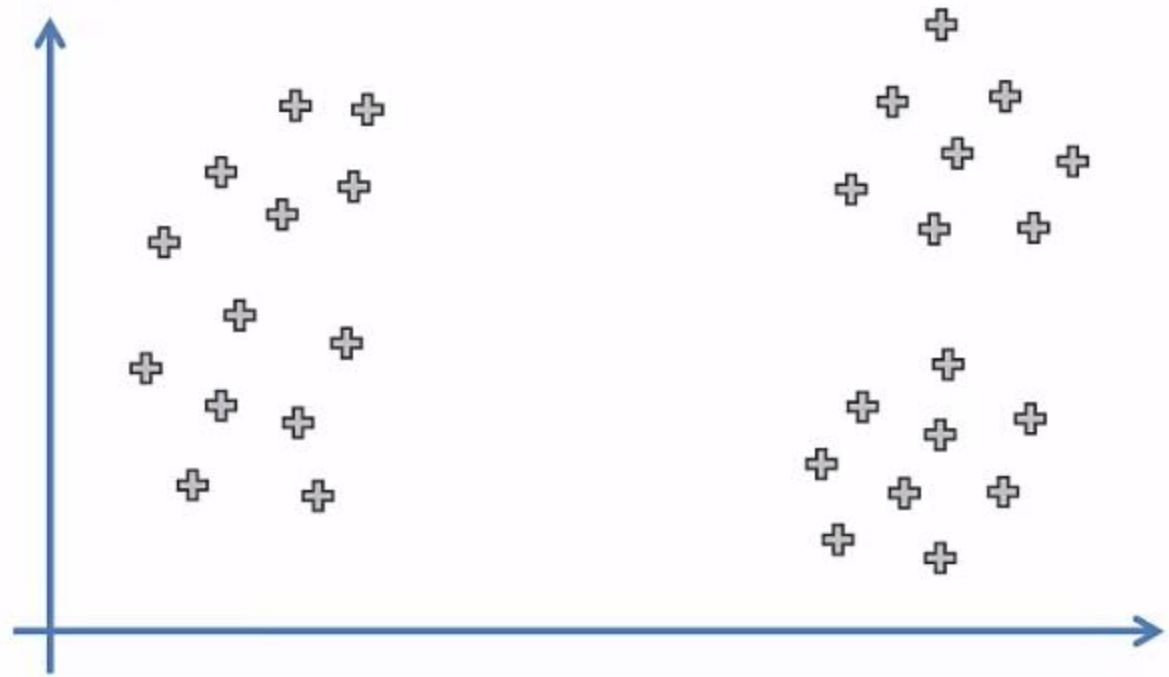
...the following three clusters



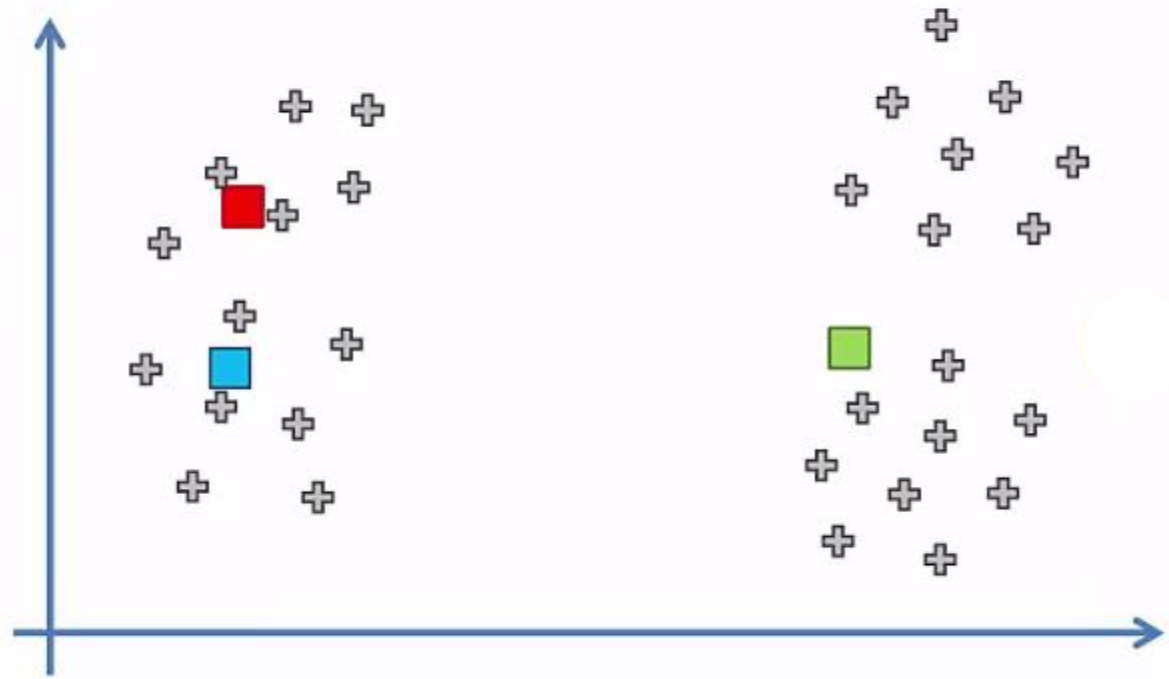
...the following three clusters

What will happen if we choose a bad random initialization?

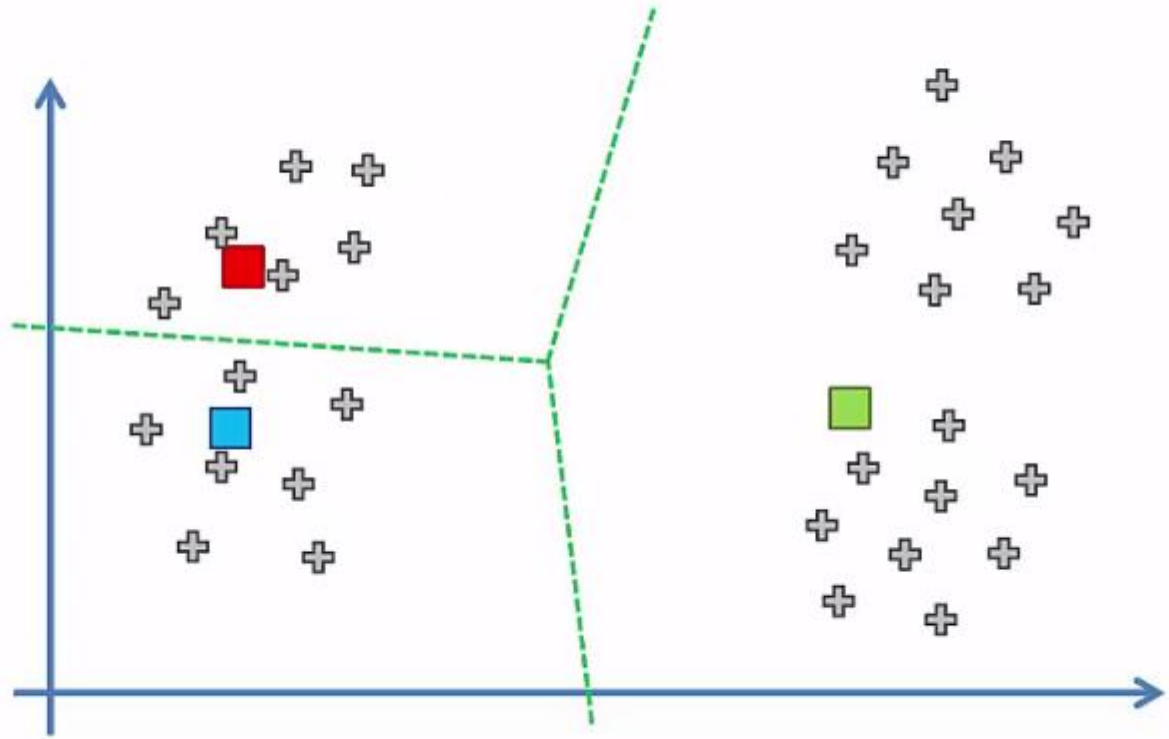
STEP 1: Choose the number K of clusters: K = 3



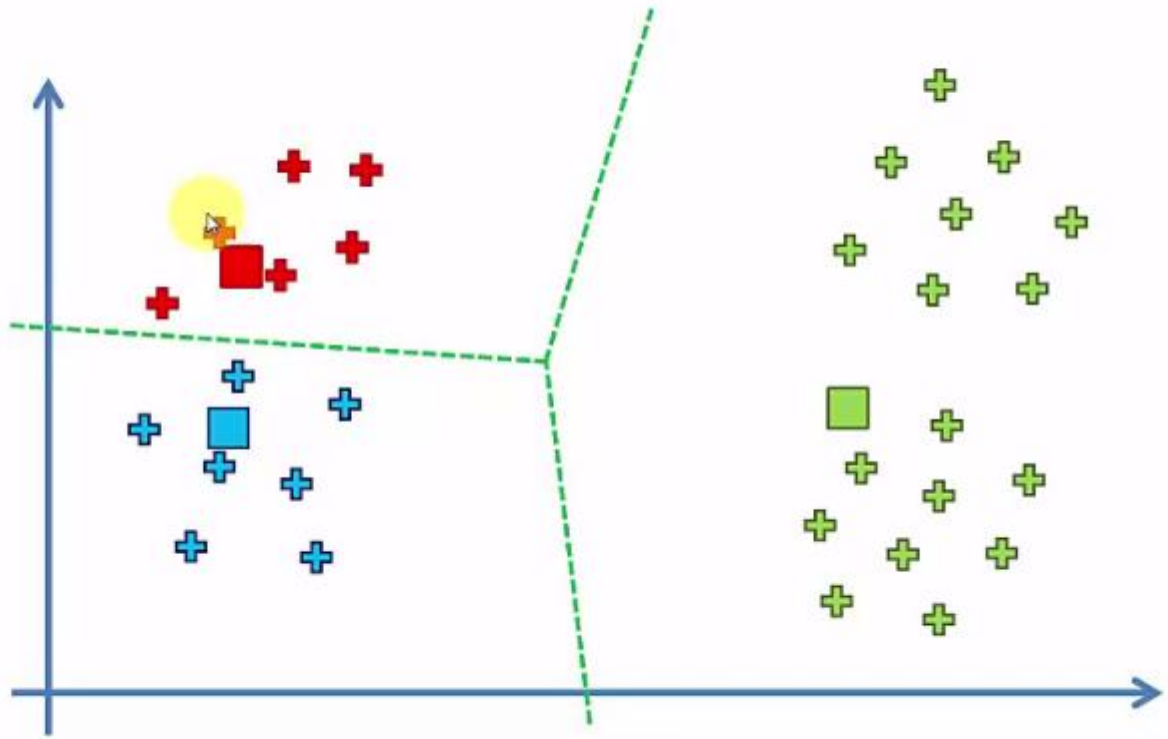
STEP 2: Select at random K points, the centroids (not necessarily from your dataset)



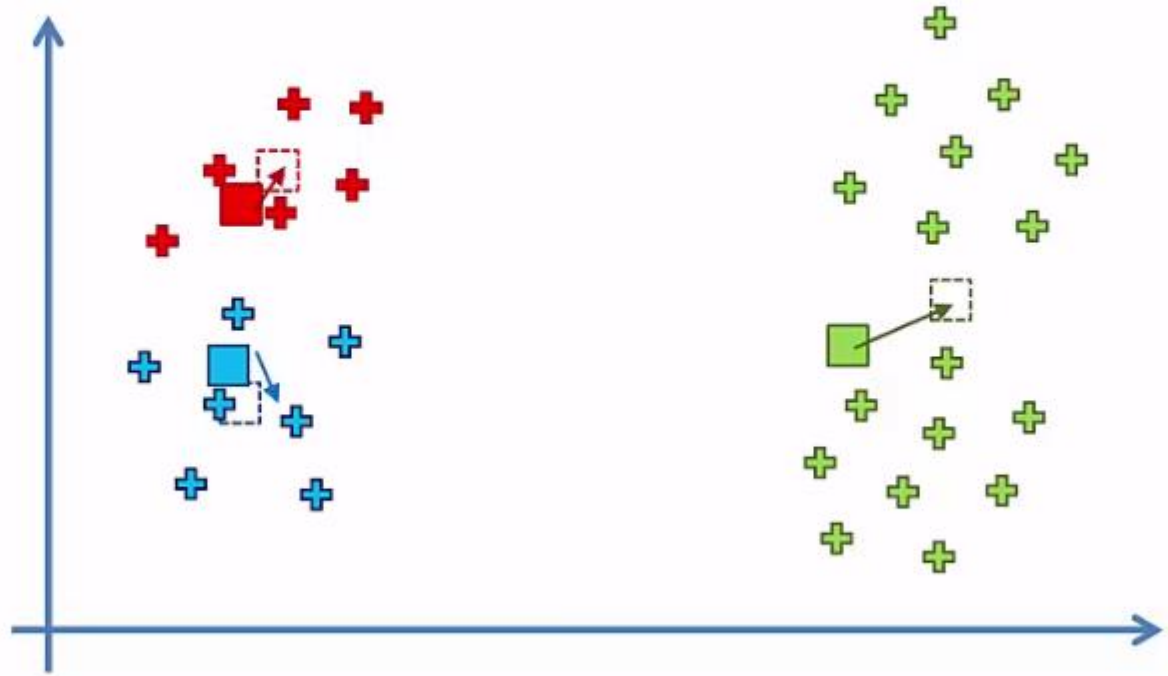
STEP 2: Select at random K points, the centroids (not necessarily from your dataset)



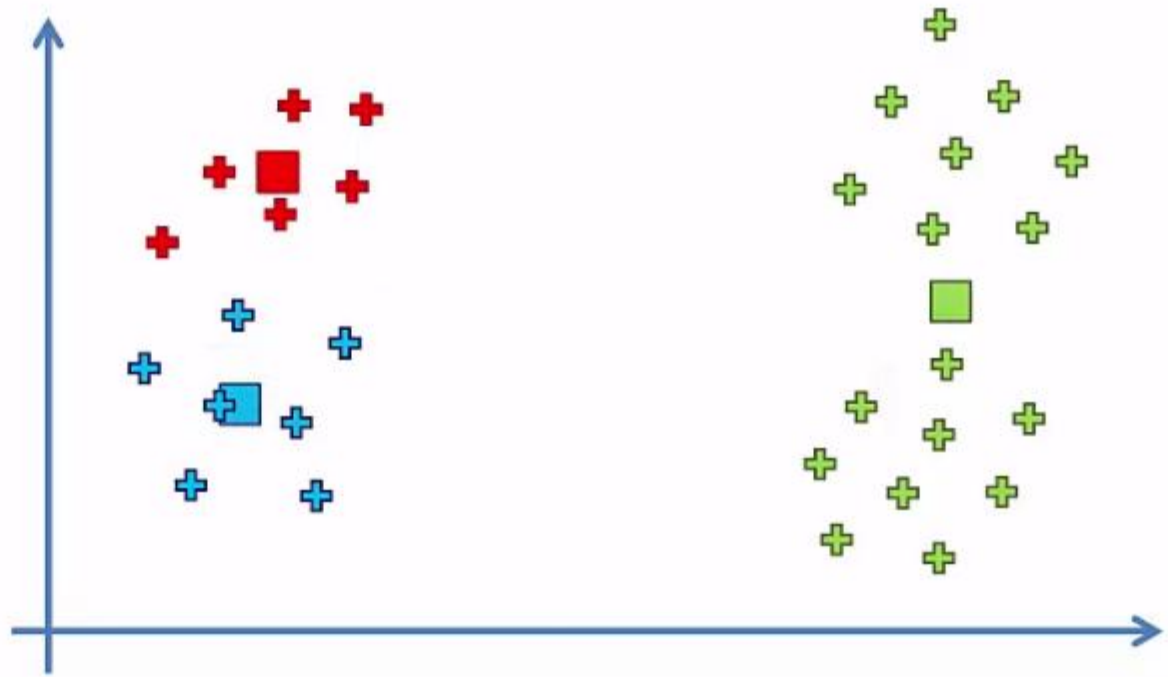
STEP 3: Assign each data point to the closest centroid ➡ That forms K clusters



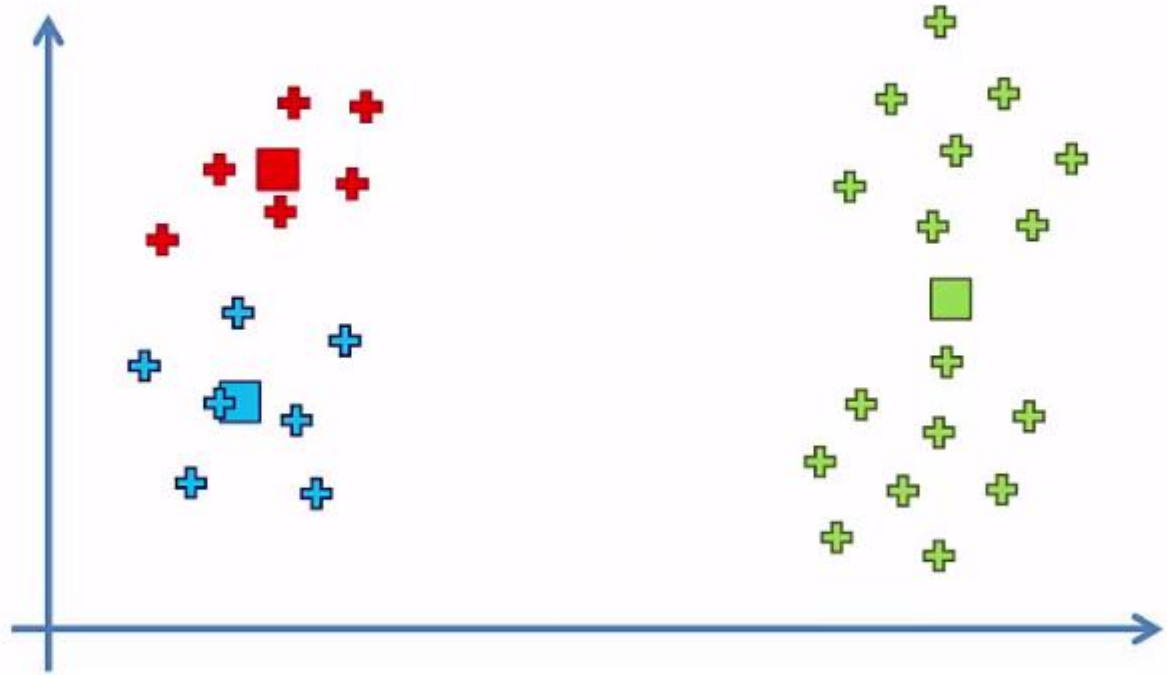
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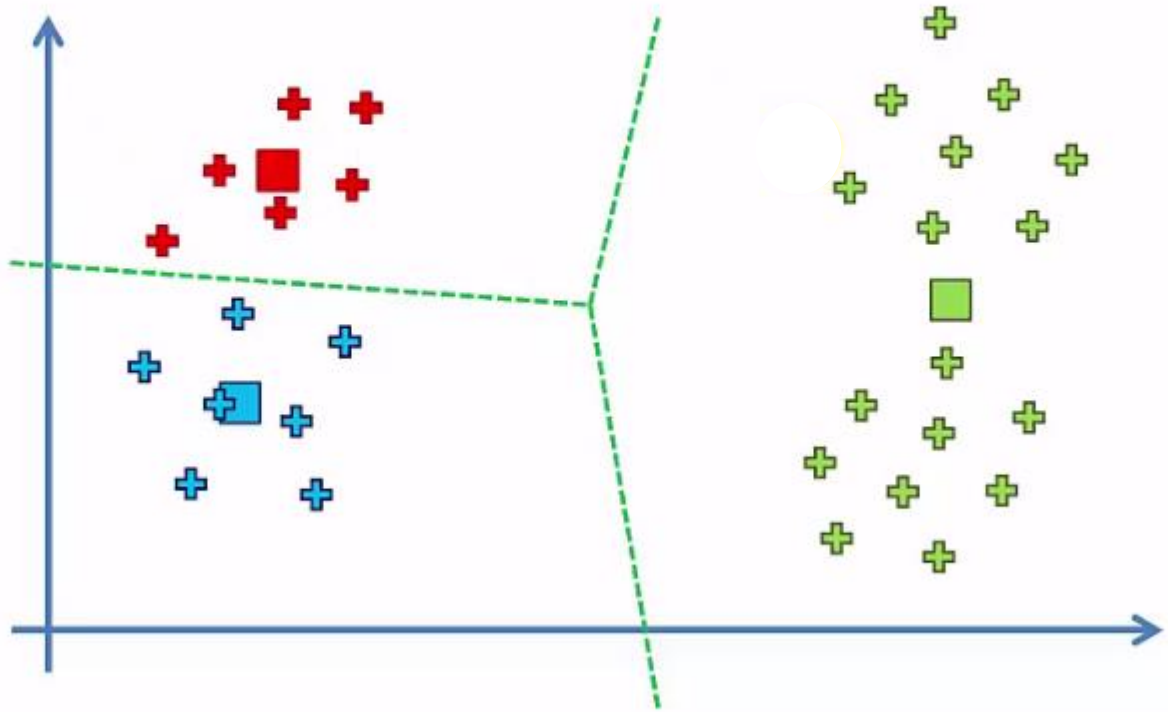
STEP 4: Compute and place the new centroid of each cluster



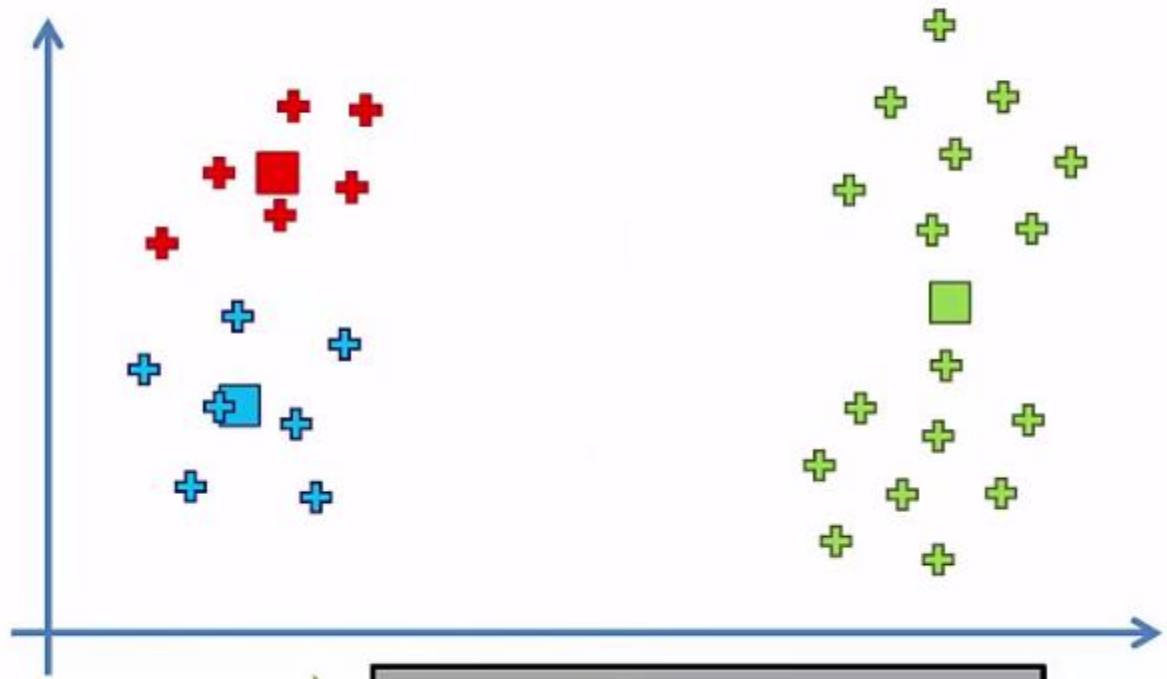
STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



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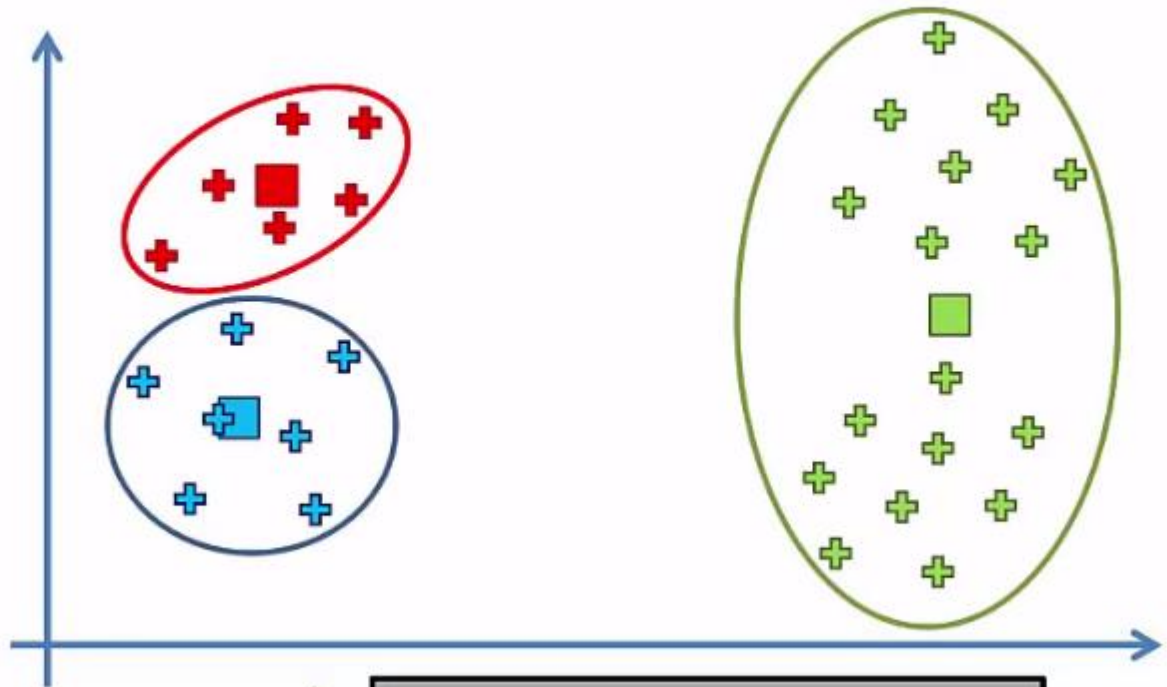


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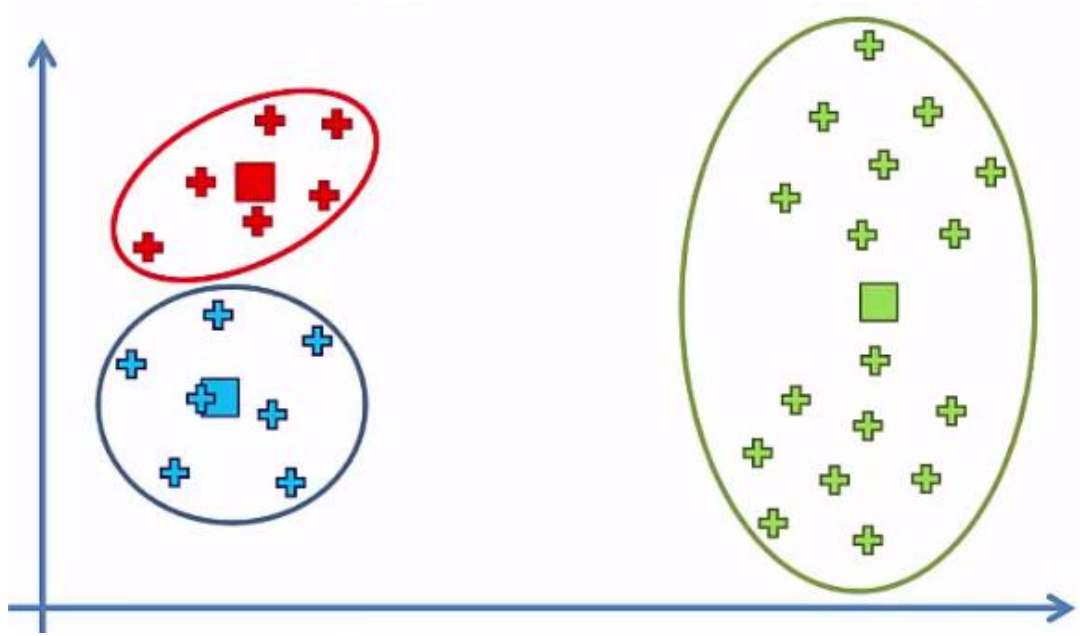


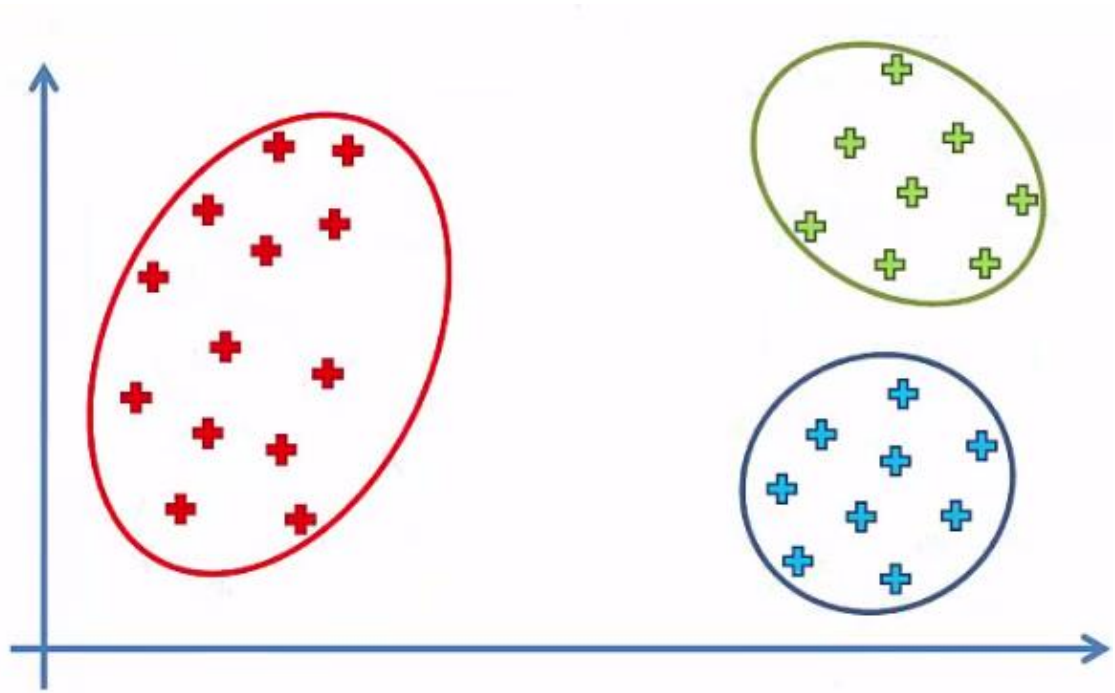
Your Model is Ready

STEP 5: Reassign each data point to the new closest centroid.
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Your Model is Ready



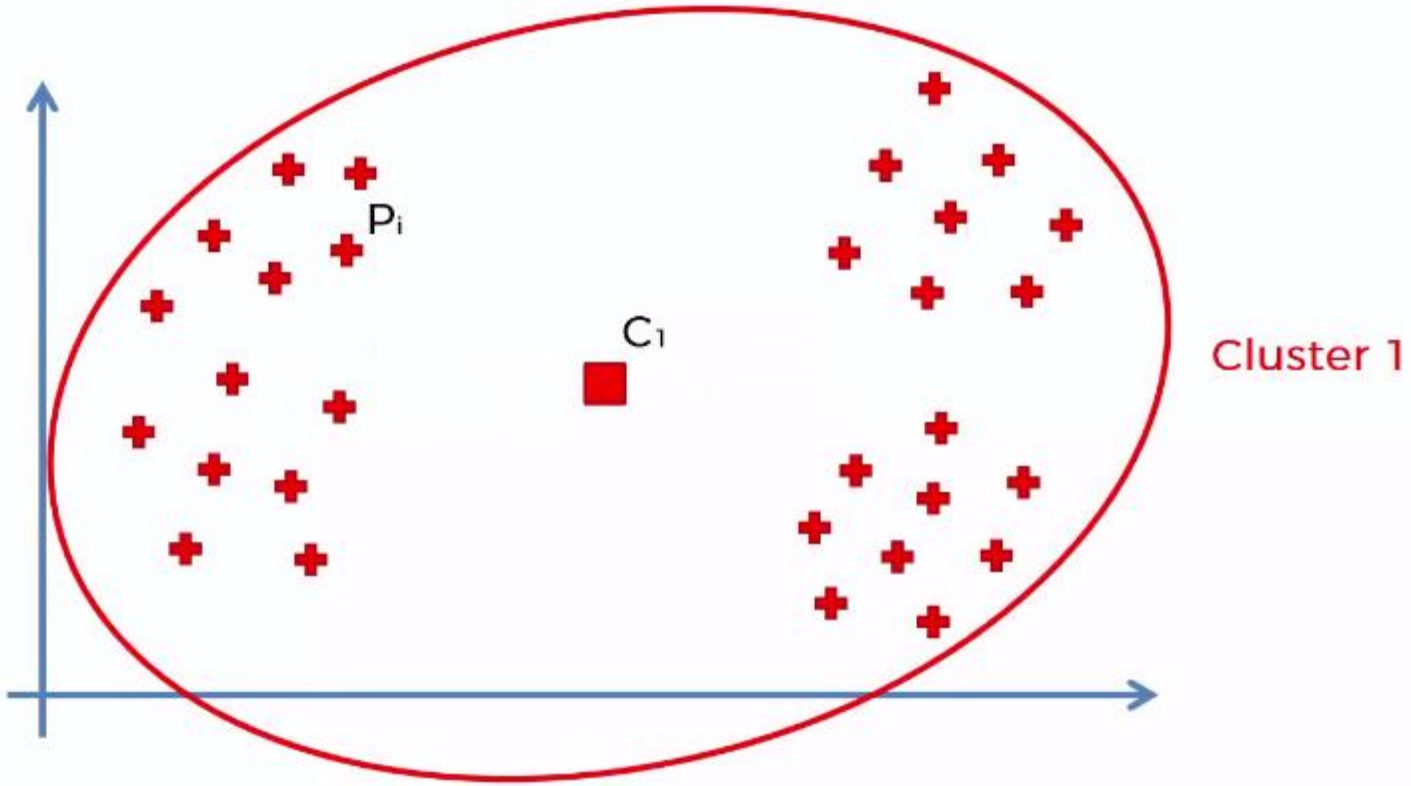



Solution
K-Means++

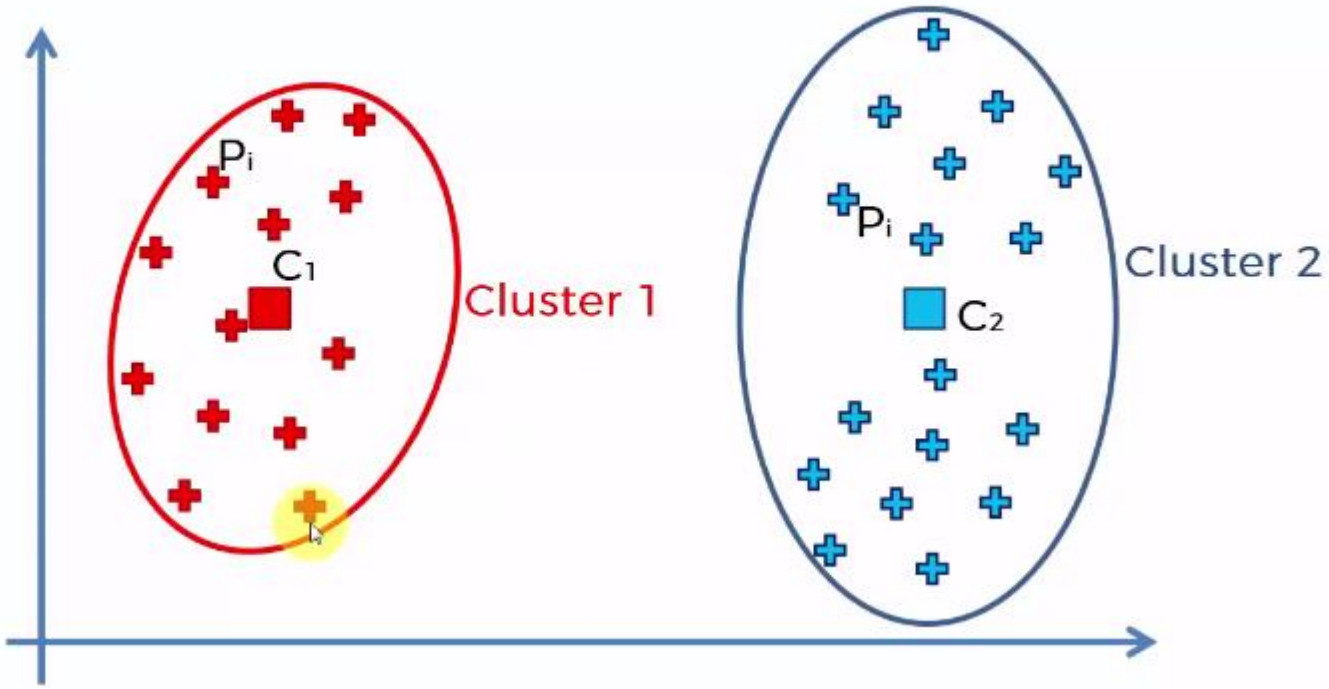
Choosing the Right Number of Clusters

→ Within Cluster Sum of Square (WCSS)

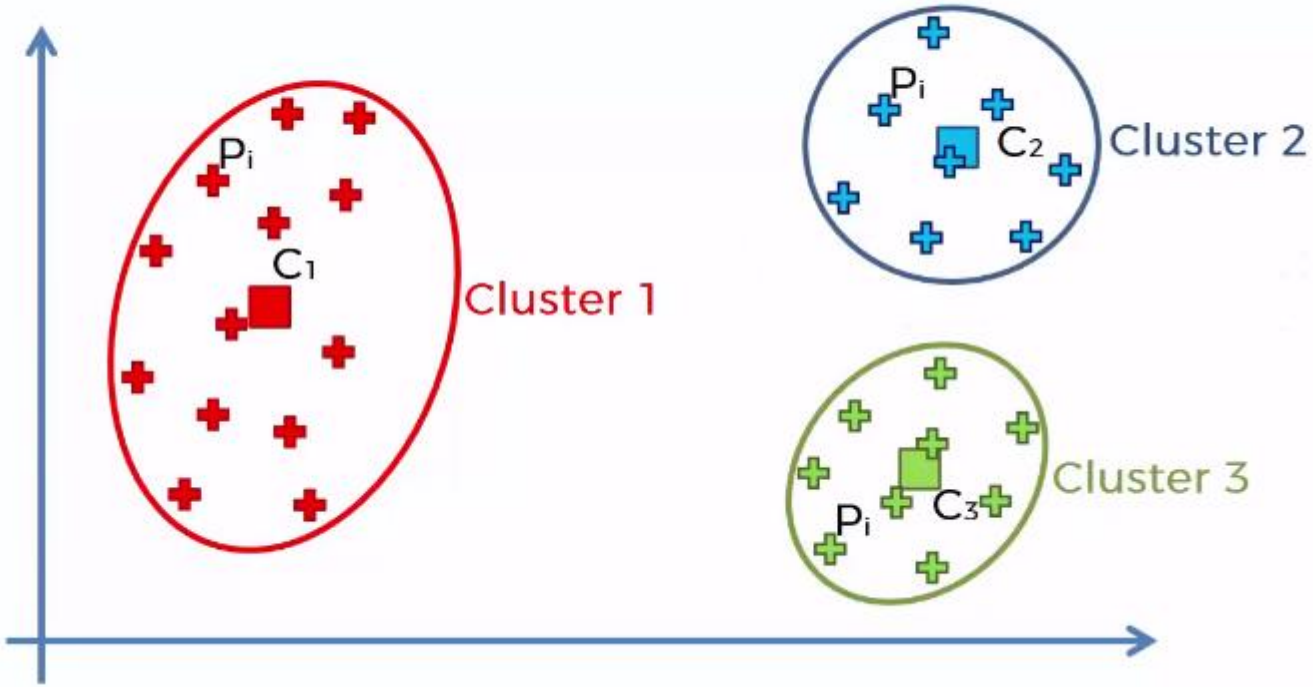
Let us consider the following scenarios…



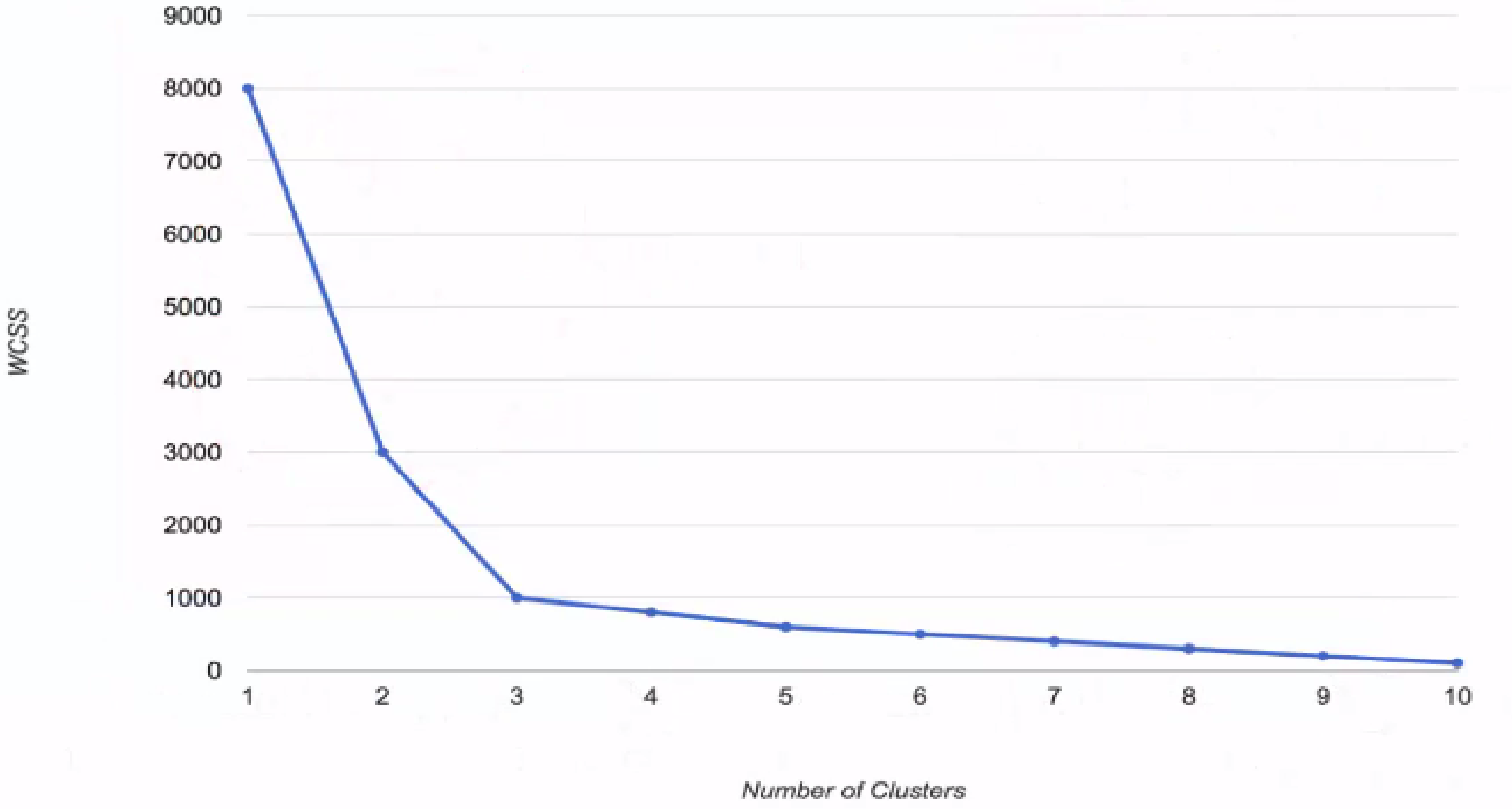
$$\mathbf{WCSS_1} = \sum_{\mathbf{p_i \text{ in cluster } 1}} \mathbf{distance}\,(\mathbf{P_i}, \mathbf{C1})^2$$



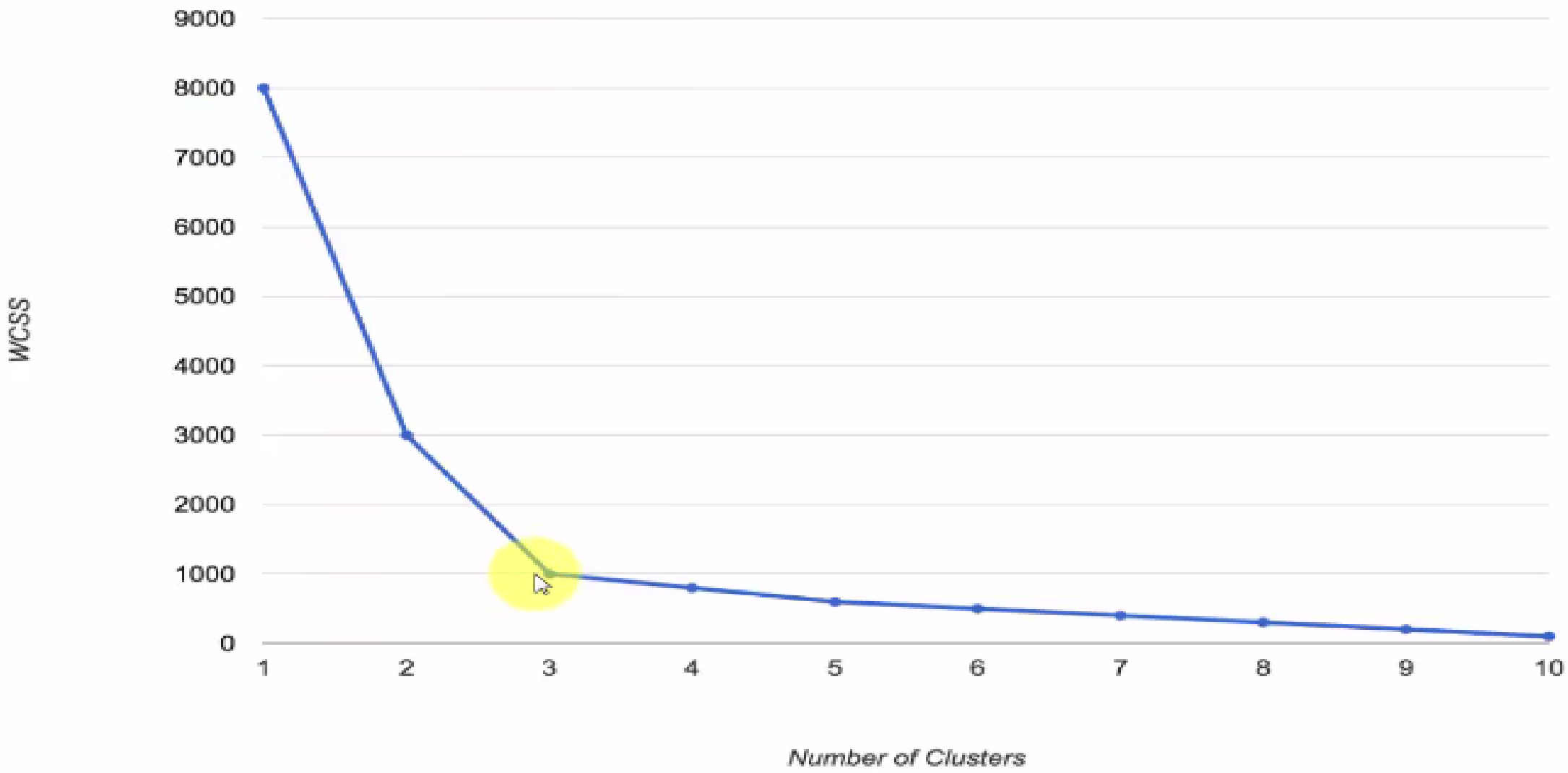
$$\text{WCSS} = \sum_{p_i \text{ in cluster 1}} \text{distance}\,(P_i, C1)^2 + \sum_{p_i \text{ in cluster 2}} \text{distance}\,(P_i, C_2)^2$$



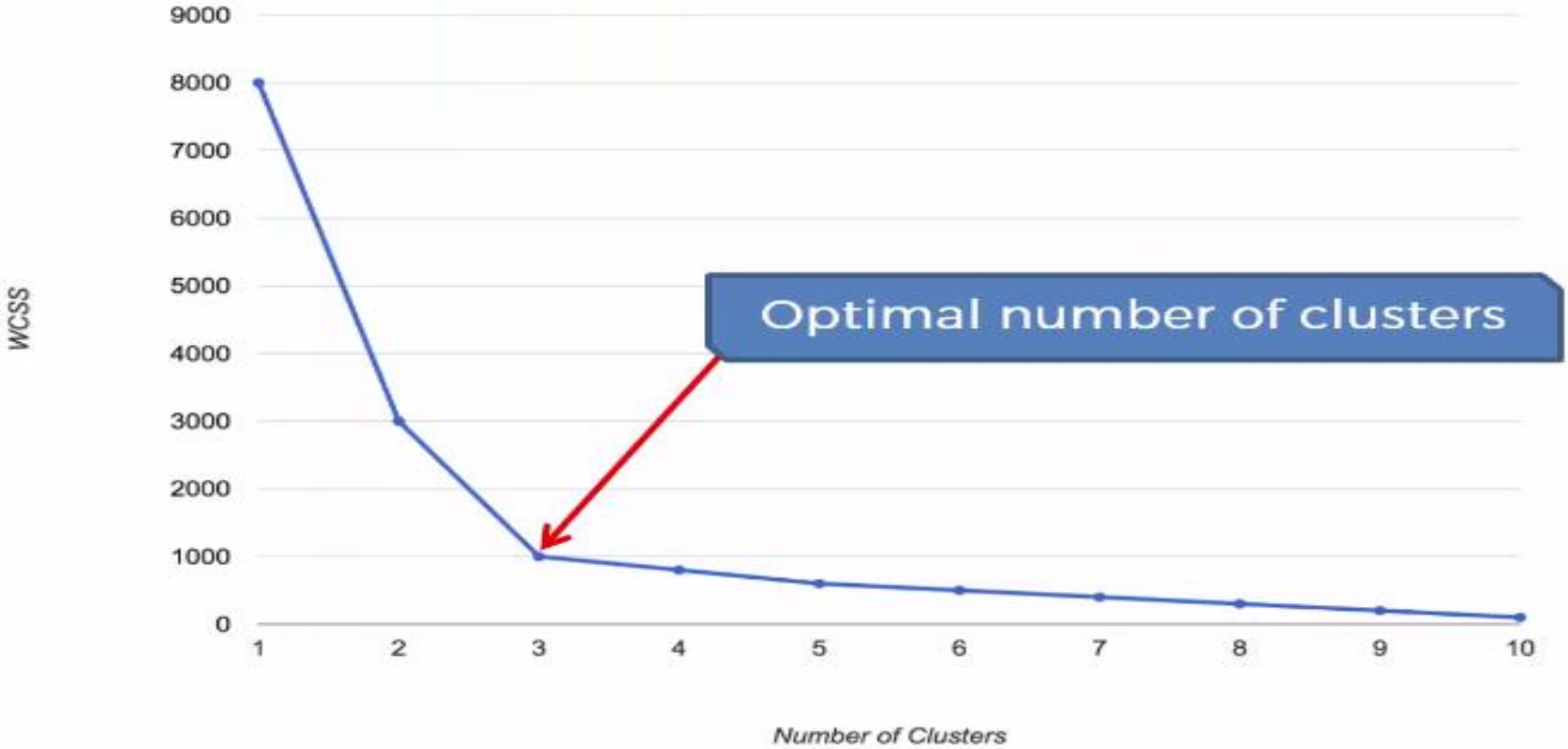
$$\text{WCSS} = \sum_{p_i \text{ in cluster 1}} \text{distance}\,(P_i, C1)^2 + \sum_{p_i \text{ in cluster 2}} \text{distance}\,(P_i, C_2)^2 + \sum_{p_i \text{ in cluster 3}} \text{distance}\,(P_i, C_3)^2$$


9000
8000
7000
6000
5000
4000
3000
2000
1000
0
WCSS
1
2
3
4
5
6
7
8
9
10
Number of Clusters

The Elbow Method



The Elbow Method



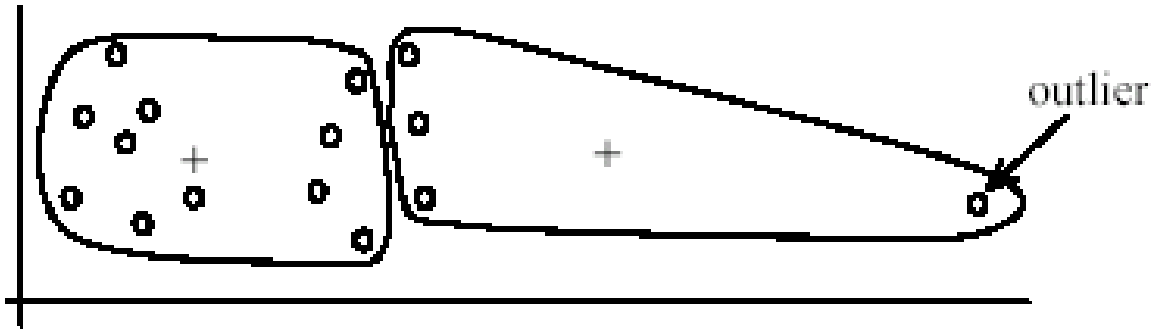
Strengths of K-means

- **Simple**: easy to understand and to implement
- **Efficient:** Time complexity: $O(tkn)$
  - n-is the number of data points
  - *k-is the number of clusters*
  - *t- is the number of iterations*
  - Since both k and t are small. k-means is considered a linear algorithm.
- K-means is the most popular clustering algorithm.

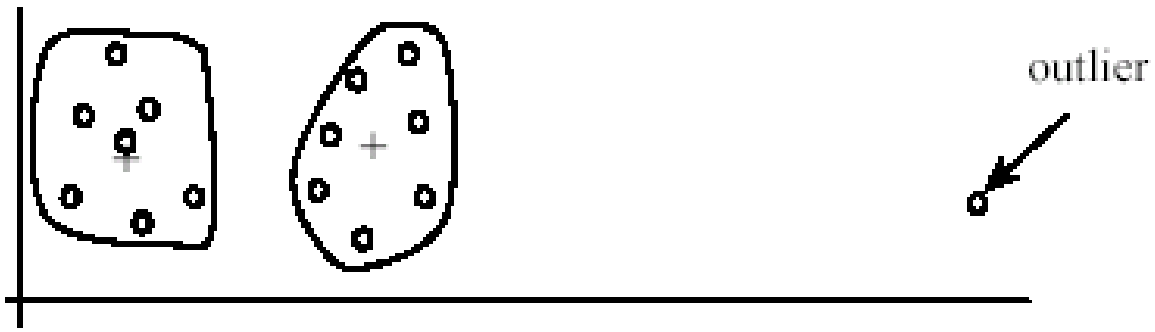
Weaknesses of K-means

- The user needs to specify number of clusters (K).
- The algorithm is sensitive to **outliers**
  - Outliers are data points that are **very far away from other data points**.
  - Outliers could be **errors** in the data recording or some **special data points** with very different values.

Outliers



(A): Undesirable clusters



(B): Ideal clusters

Handling with outliers

- Remove some data points that are much further away from the centroids
- To be safe, monitor these possible outliers over a few iterations and then decide to remove them.

Recap…

- K-means clustering can handle larger datasets
- Data points are not permanently committed to a cluster.
- They may be in different clusters during the process of clustering to improves the overall performance
- The use of means implies that all data points must be continuous
- This approach can be severely affected by outliers.